

Fourier Analysis of Wagner’s k -Tree Algorithm for SIS

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Abstract

We analyze a simplified full-pair filtering variant of Wagner’s k -tree algorithm for the Short Integer Solution problem in the infinity norm (SIS^∞). The variant deliberately enumerates all ordered pairs at every level and filters them directly by an ℓ_∞ syndrome constraint, thereby sacrificing running time in order to isolate the success-versus-cancellation question.

For this full-pair process, the empirical list distribution satisfies an exact Fourier recurrence. We derive this recurrence, introduce an affine-goodness framework for controlling the affine-coset averaging caused by conditioning, and prove a denominator lower-bound mechanism using a second-moment argument over projective lines. These tools give a rigorous recursive criterion under which the final empirical distribution has small mass at zero.

The main conclusion is negative for this Fourier strategy. Within the affine-goodness recursion developed here, the final mass condition already forces

$$m - n = \Omega\left(\frac{n \log n}{\log \log n}\right),$$

whereas the target Wagner/SIS regime is $m - n = \omega(n/\log \log n)$. Thus the paper identifies a precise obstruction to this natural Fourier-analytic route. It does not rule out Wagner’s algorithm itself, nor does it rule out Fourier analyses based on finer statistics than the one-dimensional spectral-tail quantity studied here.

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1 Introduction

1.1 Background and motivation

The Short Integer Solution problem in the infinity norm, $\text{SIS}_{n,m,q,\beta}^\infty$, asks: given a uniformly random matrix $A \in \mathbb{Z}_q^{n \times m}$, find a nonzero vector $x \in \mathbb{Z}^m$ such that

$$Ax \equiv 0 \pmod{q}, \quad \|x\|_\infty \leq \beta.$$

This problem is a central average-case lattice problem. It is closely related to the q -ary lattice

$$\Lambda_q^\perp(A) := \{x \in \mathbb{Z}^m : Ax \equiv 0 \pmod{q}\}.$$

A Wagner-style approach to SIS starts from many short vectors in the free coordinates and repeatedly combines pairs whose newly exposed syndrome block is small. In its lazy-modulus-switching form, level i uses a smaller modulus p_i for bucketing; two vectors are combined when their images under a quantized version of the current syndrome block coincide. This reduces the number of buckets from roughly q^{b_i} to roughly $p_i^{b_i}$, at the cost of introducing a rounding error of size about q/p_i in the newly processed block [1, 5, 3].

The heuristic Wagner calculation suggests subexponential list size

$$N = 2^{\Theta(n/\log \log n)}$$

for

$$q = \text{poly}(n), \quad \beta = q/\text{polylog}(n), \quad m - n = \omega(n/\log \log n).$$

However, the central success issue is that the final list might, in principle, concentrate at the zero vector. In the original generalized birthday setting this cancellation issue is avoided by using linearly independent initial vectors, but that argument is not available for the SIS parameter regime where only $m - n$ free coordinates are present. Related difficulties appear in the BKW/LWE literature: Kirchner and Fouque's subexponential claim for narrow-error LWE relied on a Wagner/BKW-style step, while Herold, Kirshanova, and May later identified a gap in the sample-amplification analysis when the number of samples is only linear in n [5, 4].

A recent work of Ducas, Engelberts, and Loyer gives a provable subexponential-time algorithm for SIS^∞ in this regime by modifying Wagner's algorithm using discrete-Gaussian sampling and randomized rounding [2]. Their result shows that a suitably modified Wagner paradigm can be made rigorous. The present paper studies a different question: how far can one push a direct Fourier analysis of the combinatorial full-pair filtering process, without introducing the discrete-Gaussian modifications?

1.2 The simplified full-pair filter

We analyze a deliberately simplified variant. At level i , instead of subsampling pairs or pairing elements only within lazy buckets, the algorithm enumerates every ordered pair of previous list positions, keeps all differences x satisfying

$$\| -A'_i x \|_{\infty, q} \leq 2^i,$$

and retains multiplicities. This full-pair filter is not intended as an efficient implementation. Its purpose is to remove pairing-order and subsampling issues so that the empirical list distribution is exactly governed by a Fourier recurrence.

This simplification is also a relaxation of a natural lazy-bucketing variant. Indeed, if the lazy bucket map at level i has bucket diameter at most 2^i in each coordinate, then every pair combined by the lazy algorithm is accepted by the full-pair filter. Hence, for a common initial list and common row blocks, the lazy intermediate list embeds into the corresponding full-pair list, with multiplicity. This formal containment is stated in Lemma 3.4. Thus a failure of the full-pair Fourier analysis is a meaningful sanity check: it shows that the present spectral method is too weak even after replacing the algorithm by a much larger and easier-to-succeed process.

1.3 Fourier program and outcome

Let $G = \mathbb{Z}_q^d$ with $d = m - n$. The level- i state is the empirical distribution μ_i obtained by choosing a uniformly random list position from the indexed list L_i . Each round consists of:

(i) *Differencing*:

$$\nu_i = \mu_{i-1} * \check{\mu}_{i-1}, \quad \widehat{\nu}_i(\omega) = |\widehat{\mu_{i-1}}(\omega)|^2.$$

(ii) *Filtering and renormalization*: multiply by

$$h_i(x) := \mathbf{1}_{C_i}(-A_i'x), \quad C_i := \{y \in \mathbb{Z}_q^{b_i} : \|y\|_{\infty, q} \leq 2^i\},$$

then normalize.

The resulting exact recurrence is

$$\widehat{\mu}_i(\omega) = \frac{\sum_{\tau \in \mathbb{Z}_q^{b_i}} K_i(\tau) |\widehat{\mu_{i-1}}(\omega + A_i'^T \tau)|^2}{\sum_{\tau \in \mathbb{Z}_q^{b_i}} K_i(\tau) |\widehat{\mu_{i-1}}(A_i'^T \tau)|^2},$$

where

$$K_i(\tau) = \prod_{j=1}^{b_i} D_{2^i}(\tau_j)$$

is a product Dirichlet kernel.

The recurrence shows the main difficulty: conditioning is not a pointwise Fourier multiplier. Instead, each output frequency averages the previous spectrum over the affine coset $\omega + \text{im}(A_i'^T)$ with signed Dirichlet weights. We introduce an affine-goodness framework adapted to this coset averaging and prove a denominator concentration bound using a projective-line second-moment decomposition.

The final result is an obstruction. The affine-goodness recursion forces too much cumulative bad-mass amplification: maintaining thresholds below 1 across all $r = \Theta(\log n)$ levels and imposing final mass at zero at most $1/2$ requires

$$d = m - n = \Omega\left(\frac{n \log n}{\log \log n}\right).$$

Therefore this method does not prove success in the intended regime $d = \omega(n/\log \log n)$. The obstruction is specific to the recursive spectral-tail analysis developed here; it is not a lower bound against Wagner's algorithm.

2 Preliminaries

2.1 Notation

Throughout, q is a prime with $q \rightarrow \infty$ (concretely, $q = \text{poly}(n)$), and d, n, m are positive integers with $d := m - n$. All arithmetic in $\mathbb{Z}_q$ is modulo q . All logarithms are natural unless a base is explicitly indicated; in particular, $\log_2$ denotes base-2 logarithm.

Definition 2.1 (Centered representative and norms). For $y \in \mathbb{Z}_q$, the *centered representative* is the unique integer $\bar{y} \in \{-(q-1)/2, \dots, (q-1)/2\}$ with $\bar{y} \equiv y \pmod{q}$. Define $|y|_q := |\bar{y}|$. For $y = (y_1, \dots, y_b) \in \mathbb{Z}_q^b$, set $\|y\|_{\infty, q} := \max_{1 \leq j \leq b} |y_j|_q$.

2.2 Fourier analysis on $\mathbb{Z}_q^d$

Definition 2.2 (Characters). For $\omega \in \mathbb{Z}_q^d$, define the character $\chi_\omega : \mathbb{Z}_q^d \rightarrow \mathbb{C}$ by

$$\chi_\omega(x) := \exp\left(\frac{2\pi i}{q} \langle \omega, x \rangle\right), \quad \langle \omega, x \rangle := \sum_{j=1}^d \omega_j x_j \pmod{q}.$$

The characters satisfy the orthogonality relation

$$\sum_{x \in \mathbb{Z}_q^d} \chi_\eta(x) = \begin{cases} q^d, & \eta = 0, \\ 0, & \eta \neq 0. \end{cases} \quad (1)$$

We use two Fourier-transform conventions, distinguished by context.

Definition 2.3 (Fourier transforms). Let $f : \mathbb{Z}_q^d \rightarrow \mathbb{C}$.

(a) The *unnormalized Fourier transform* is

$$\hat{f}(\omega) := \sum_{x \in \mathbb{Z}_q^d} f(x) \chi_\omega(x).$$

(b) If μ is a probability distribution on $\mathbb{Z}_q^d$ (i.e. $\mu \geq 0$, $\sum_x \mu(x) = 1$), its Fourier transform coincides with the characteristic function:

$$\hat{\mu}(\omega) = \sum_{x \in \mathbb{Z}_q^d} \mu(x) \chi_\omega(x) = \mathbb{E}_{X \sim \mu}[\chi_\omega(X)].$$

In particular, $\hat{\mu}(0) = 1$ and $|\hat{\mu}(\omega)| \leq 1$ for all ω .

The Fourier inversion formula is

$$f(x) = \frac{1}{q^d} \sum_{\omega \in \mathbb{Z}_q^d} \hat{f}(\omega) \chi_{-\omega}(x). \quad (2)$$

Proposition 2.4 (Parseval identity). For any $f : \mathbb{Z}_q^d \rightarrow \mathbb{C}$,

$$\sum_{x \in \mathbb{Z}_q^d} |f(x)|^2 = \frac{1}{q^d} \sum_{\omega \in \mathbb{Z}_q^d} |\hat{f}(\omega)|^2.$$

In particular, if μ is a probability distribution, then $\sum_x \mu(x)^2 = q^{-d} \sum_\omega |\hat{\mu}(\omega)|^2$.

Definition 2.5 (Convolution). For $f, g : \mathbb{Z}_q^d \rightarrow \mathbb{C}$, define

$$(f * g)(x) := \sum_{u \in \mathbb{Z}_q^d} f(u) g(x - u).$$

Proposition 2.6 (Convolution theorem). $\widehat{f * g}(\omega) = \widehat{f}(\omega) \cdot \widehat{g}(\omega)$ for all $\omega \in \mathbb{Z}_q^d$.

Proof. Expand $(f * g)(x) = \sum_u f(u)g(x - u)$, multiply by $\chi_\omega(x)$, and sum over x . Using $\chi_\omega(x) = \chi_\omega(u)\chi_\omega(x - u)$ and interchanging the finite sums gives the result. $\square$

Lemma 2.7 (Reflection and modulus-square). Let μ be a probability distribution on $\mathbb{Z}_q^d$. Define $\check{\mu}(x) := \mu(-x)$. Then $\widehat{\check{\mu}}(\omega) = \overline{\widehat{\mu}(\omega)}$, and consequently

$$\widehat{\mu * \check{\mu}}(\omega) = |\widehat{\mu}(\omega)|^2.$$

If in addition μ is even, i.e. $\mu(x) = \mu(-x)$ for all x , then $\widehat{\mu}(\omega) \in \mathbb{R}$ for all ω , and $\widehat{\mu * \check{\mu}}(\omega) = \widehat{\mu}(\omega)^2$.

Proof. For the first claim, substitute $y = -x$ in the definition of $\widehat{\mu}(\omega)$ and use $\chi_\omega(-y) = \overline{\chi_\omega(y)}$. For evenness, pair x and $-x$ in the Fourier sum: the sine terms cancel since $\mu(x) = \mu(-x)$ and $\sin$ is odd, leaving $\widehat{\mu}(\omega) \in \mathbb{R}$. $\square$

Definition 2.8 (Spectral tail density). For a probability distribution μ on $\mathbb{Z}_q^d$ and a threshold $\theta \in (0, 1)$, define

$$\delta(\mu, \theta) := \frac{1}{q^d} \# \{ \omega \in \mathbb{Z}_q^d : |\widehat{\mu}(\omega)| > \theta \}.$$

When $\mu = \mu_i$ is clear from context, we write $\delta_i(\theta) := \delta(\mu_i, \theta)$.

2.3 The Dirichlet kernel

Definition 2.9. For an integer $B \geq 0$ and $a \in \mathbb{Z}_q$, define

$$D_B(a) := \sum_{s=-B}^B e^{2\pi i a s / q}.$$

Lemma 2.10 (Basic properties). Let $1 \leq B < q/2$.

$$(i) \quad D_B(0) = 2B + 1.$$

$$(ii) \quad \text{For } a \not\equiv 0 \pmod{q},$$

$$D_B(a) = \frac{\sin((2B + 1)\pi a / q)}{\sin(\pi a / q)}. \quad (3)$$

$$(iii) \quad D_B(-a) = D_B(a) \text{ for all } a \in \mathbb{Z}_q.$$

$$(iv) \quad |D_B(a)| \leq 2B + 1 \text{ for all } a \in \mathbb{Z}_q.$$

$$(v) \quad \text{For } 1 \leq |a|_q \leq q/2,$$

$$|D_B(a)| \leq \frac{q}{2|a|_q}. \quad (4)$$

Proof. Part (i) is immediate. For (ii), sum the geometric series $\sum_{s=-B}^B (e^{2\pi i a/q})^s$ and simplify. Part (iii) follows because

$$D_B(a) = 1 + 2 \sum_{s=1}^B \cos(2\pi a s/q),$$

which is an even function of a . Part (iv) is the triangle inequality. For (v), use $\sin(\pi a/q) \geq 2|a|_q/q$ (valid for $|a|_q \leq q/2$ since $\sin x \geq 2x/\pi$ on $[0, \pi/2]$) and $|\sin(\cdot)| \leq 1$ in the numerator of (3). $\square$

Remark 2.11. The product kernel at level i is $K_i(\tau) = \prod_{j=1}^{b_i} D_{2^i}(\tau_j)$ for $\tau \in H_i = \mathbb{Z}_q^{b_i}$. By Lemma 2.10(iii), K_i is an even function on H_i , and $K_i(0) = W_i^{b_i} = |C_i|$. The kernel ℓ^1 -mass factors as $S_i = \sum_{\tau \in H_i} |K_i(\tau)| = T_q(W_i)^{b_i}$, where $T_q(W) := \sum_{a \in \mathbb{Z}_q} |D_{(W-1)/2}(a)|$.

3 The k -Tree Algorithm for SIS

3.1 The SIS^∞ problem and systematic form

Definition 3.1 ($\text{SIS}_{n,m,q,\beta}^\infty$). Let $n, m, q \in \mathbb{N}$ with $m > n$, and let $\beta > 0$. Given $A \in \mathbb{Z}_q^{n \times m}$, the problem $\text{SIS}_{n,m,q,\beta}^\infty$ is to find a nonzero vector $x \in \mathbb{Z}^m$ such that

$$Ax \equiv 0 \pmod{q}, \quad \|x\|_\infty \leq \beta.$$

We condition on the event that A has full row rank over $\mathbb{F}_q$. For a uniform random $A \in \mathbb{F}_q^{n \times m}$, this event has probability

$$\prod_{j=0}^{n-1} (1 - q^{j-m}),$$

and hence the failure probability is at most

$$\sum_{j=0}^{n-1} q^{j-m} \leq \frac{q^{n-m}}{1 - 1/q}.$$

Thus, when $d := m - n \rightarrow \infty$, the conditioning loses only negligible probability.

On the full-rank event, some set of n columns is invertible. Permuting columns and left-multiplying by the inverse of this column submatrix puts the matrix in systematic form

$$A = [A' \mid I_n], \quad A' \in \mathbb{Z}_q^{n \times d}, \quad d = m - n.$$

Column permutation preserves the ℓ_∞ norm, and left multiplication by an invertible matrix over $\mathbb{F}_q$ preserves the solution set of the homogeneous system. Therefore this reduction is without loss of generality for SIS^∞ .

In systematic form, every vector $z \in \mathbb{Z}_q^d$ gives a syndrome-completing vector

$$x(z) := (z, -A'z \bmod q) \in \mathbb{Z}_q^m,$$

and

$$Ax(z) = A'z - A'z \equiv 0 \pmod{q}.$$

Thus it suffices to find a nonzero $z \in \mathbb{Z}_q^d$ such that the centered representative of $x(z)$ has small infinity norm.

3.2 Algorithm description

The matrix A' is partitioned row-wise into r blocks. Fix block sizes $(b_i)_{i=1}^r$ with $\sum_{i=1}^r b_i = n$, and set $n_0 := 0$, $n_i := \sum_{j=1}^i b_j$. Let $A'_i \in \mathbb{Z}_q^{b_i \times d}$ denote rows $n_{i-1} + 1$ through n_i of A' .

The algorithm maintains indexed lists of vectors in $G = \mathbb{Z}_q^d$. The word “list” is used in the multiplicity-preserving sense: two equal vectors appearing at two different list positions are treated as two distinct entries.

Definition 3.2 (Indexed lists and empirical laws). Let $L = (x_s)_{s \in I}$ be a finite indexed list with entries in G , where I is a finite index set. Its size is $|L| := |I|$, counted with multiplicity. The empirical distribution associated with L is

$$\mu_L(z) := \frac{1}{|L|} \#\{s \in I : x_s = z\}, \quad z \in G.$$

Equivalently, sampling from μ_L means choosing a uniformly random list position $s \in I$ and outputting x_s .

At each level, the simplified algorithm enumerates every ordered pair of previous-level list positions exactly once and keeps all accepted differences. The acceptance cube at level i is

$$C_i := \{y \in \mathbb{Z}_q^{b_i} : \|y\|_{\infty, q} \leq 2^i\}.$$

Algorithm (Simplified full-pair k -Tree for SIS)

Input: $A' \in \mathbb{Z}_q^{n \times d}$; parameters $N, r, (b_i)_{i=1}^r$.

Output: A nonzero $x \in \mathbb{Z}_q^d$ such that, after taking centered representatives,

$$\|(x, -A'x \bmod q)\|_{\infty, q} \leq 2^r,$$

or FAILURE.

1. **Initialize.** Sample an indexed list

$$L_0 = (X_1^{(0)}, \dots, X_{N_0}^{(0)}), \quad N_0 := 3^r N,$$

where the entries are independent samples from $\text{Unif}(\{-1, 0, 1\}^d)$.

2. **For** $i = 1, \dots, r$: Let $N_{i-1} := |L_{i-1}|$ and write

$$L_{i-1} = (X_s^{(i-1)})_{s \in [N_{i-1}]}.$$

Form the indexed virtual list

$$\widehat{L}_i := (X_s^{(i-1)} - X_t^{(i-1)})_{(s,t) \in [N_{i-1}]^2}.$$

Thus $\widehat{L}_i$ is indexed by ordered pairs and has length N_{i-1}^2 , counted with multiplicity.

Define the accepted index set

$$I_i := \{(s, t) \in [N_{i-1}]^2 : \|-A'_i(X_s^{(i-1)} - X_t^{(i-1)}) \bmod q\|_{\infty, q} \leq 2^i\}.$$

Set

$$L_i := (X_s^{(i-1)} - X_t^{(i-1)})_{(s,t) \in I_i},$$

with any fixed deterministic ordering of the accepted ordered pairs.

3. **Output.** If L_r contains a nonzero vector x , output $(x, -A'x \bmod q)$, interpreted via centered representatives; otherwise output FAILURE.

Definition 3.3 (Diameter-controlled lazy bucket maps). Let $p \leq q$ be positive integers. A bucket map

$$Q_{p,q} : \mathbb{Z}_q \rightarrow [p]$$

is called diameter-controlled if every fiber has centered diameter at most $\lceil q/p \rceil$, i.e.,

$$Q_{p,q}(u) = Q_{p,q}(v) \implies |u - v|_q \leq \lceil q/p \rceil.$$

Such maps exist by partitioning the centered representatives $\{-(q-1)/2, \dots, (q-1)/2\}$ into p consecutive intervals of lengths at most $\lceil q/p \rceil$.

For vectors in $\mathbb{Z}_q^b$, we apply $Q_{p,q}$ coordinatewise.

Lemma 3.4 (Full-pair filter contains lazy-bucket outputs). *Fix the same initial indexed list L_0 and the same row blocks $A'_1, \dots, A'_r$. Consider a lazy-bucketing algorithm that, at level i , uses*

$$p_i := \left\lceil \frac{q}{2^i} \right\rceil$$

and a diameter-controlled bucket map $Q_{p_i,q}$ applied coordinatewise to $-A'_i x \in \mathbb{Z}_q^{b_i}$. Suppose this lazy algorithm combines only pairs $a, b \in L_{i-1}^{\text{lazy}}$ satisfying

$$Q_{p_i,q}^{\otimes b_i}(-A'_i a) = Q_{p_i,q}^{\otimes b_i}(-A'_i b).$$

Let L_i^{fp} denote the full-pair filter list that enumerates all ordered pairs from L_{i-1}^{fp} and accepts every difference x with

$$\| -A'_i x \|_{\infty,q} \leq 2^i.$$

Then, for every $i = 0, \dots, r$, every lazy output entry at level i also appears as an entry of the full-pair list at level i , counted with multiplicity. Equivalently, there is an injection from lazy list positions into full-pair list positions preserving the stored vector.

Proof. We prove the claim by induction on i . The case $i = 0$ is immediate because the initial indexed lists are identical.

Assume the claim holds at level $i - 1$. Let $x = a - b$ be an entry produced by the lazy algorithm at level i , where $a, b \in L_{i-1}^{\text{lazy}}$ are the two combined entries. By the induction hypothesis, the corresponding entries a, b occur in L_{i-1}^{fp} . Since the full-pair algorithm enumerates every ordered pair of previous-level positions, it enumerates this ordered pair.

It remains to check that the full-pair filter accepts $a - b$. For every coordinate $j \in [b_i]$, the lazy bucket equality implies

$$Q_{p_i,q}((-A'_i a)_j) = Q_{p_i,q}((-A'_i b)_j).$$

By the diameter property,

$$|(-A'_i a)_j - (-A'_i b)_j|_q \leq \left\lceil \frac{q}{p_i} \right\rceil.$$

Since $p_i = \lceil q/2^i \rceil$, we have $q/p_i \leq 2^i$, and hence $\lceil q/p_i \rceil \leq 2^i$. Therefore

$$\| -A'_i(a - b) \|_{\infty,q} \leq 2^i.$$

Thus the full-pair filter accepts the same difference. This proves the inductive step. $\square$

Remark 3.5 (Why this simplification is useful). The full-pair process is generally much larger than a practical lazy-bucketing implementation, because it keeps all accepted ordered-pair differences rather than choosing a small set of disjoint pairs. The point of the simplification is not efficiency. The point is that it turns the list evolution into an exact empirical distributional recurrence: choosing a uniformly random output list position is exactly the same as choosing a uniformly random ordered pair from the previous list and conditioning on acceptance. Thus the full-pair process is a natural first sanity check for any proposed success analysis.

Lemma 3.6 (Norm invariant for the full-pair filter). *Let L_i be the indexed lists produced by the simplified full-pair filter. For every $i \in \{0, \dots, r\}$ and every $x \in L_i$, the centered representative of $x \in \mathbb{Z}_q^d$ satisfies*

$$\|x\|_{\infty,q} \leq 2^i.$$

Moreover, for every processed block $j \leq i$,

$$\| -A'_j x \|_{\infty,q} \leq 2^i.$$

Consequently, every $x \in L_r$ satisfies

$$\|(x, -A'x \bmod q)\|_{\infty,q} \leq 2^r.$$

Proof. We prove the two displayed invariants simultaneously by induction on i .

For $i = 0$, every entry of L_0 lies in $\{-1, 0, 1\}^d$, so $\|x\|_{\infty, q} \leq 1 = 2^0$. There are no processed syndrome blocks.

Assume the claim holds at level $i-1$. Let $x \in L_i$. By construction, $x = a - b$ for some $a, b \in L_{i-1}$. For the free coordinates, the induction hypothesis gives

$$\|a\|_{\infty, q} \leq 2^{i-1}, \quad \|b\|_{\infty, q} \leq 2^{i-1}.$$

Choosing centered representatives coordinatewise, the representative of $a - b$ has infinity norm at most 2^i . Hence $\|x\|_{\infty, q} \leq 2^i$.

Now fix a processed block $j \leq i$. If $j = i$, then the level- i filter itself gives

$$\| -A'_i x \|_{\infty, q} \leq 2^i.$$

If $j < i$, then by the induction hypothesis,

$$\| -A'_j a \|_{\infty, q} \leq 2^{i-1}, \quad \| -A'_j b \|_{\infty, q} \leq 2^{i-1}.$$

Since

$$-A'_j x = -A'_j(a - b) = (-A'_j a) - (-A'_j b),$$

the same coordinatewise triangle inequality gives

$$\| -A'_j x \|_{\infty, q} \leq 2^i.$$

This completes the induction.

At level r , all row blocks $j = 1, \dots, r$ have been processed, and $-A'x$ is their concatenation. Combining the free-coordinate bound with the blockwise syndrome bounds gives

$$\|(x, -A'x \bmod q)\|_{\infty, q} \leq 2^r.$$

□

Lemma 3.7 (Exact empirical transition). *Fix a level $i \geq 1$, and condition on the previous indexed list*

$$L_{i-1} = (x_s)_{s \in [N_{i-1}]}.$$

Let $\mu_{i-1} := \mu_{L_{i-1}}$ be its empirical distribution. Define

$$h_i(x) := \mathbf{1}_{C_i}(-A'_i x).$$

Let X, Y be independent samples from μ_{i-1} . If

$$\alpha_i := \Pr[h_i(X - Y) = 1],$$

then $\alpha_i > 0$, and the empirical distribution $\mu_i := \mu_{L_i}$ of the next list satisfies

$$\mu_i(z) = \Pr[X - Y = z \mid h_i(X - Y) = 1] = \frac{\Pr[X - Y = z, h_i(X - Y) = 1]}{\alpha_i}.$$

Equivalently,

$$\mu_i = \frac{(\mu_{i-1} * \check{\mu}_{i-1}) \cdot h_i}{\alpha_i}.$$

Proof. For each $z \in G$, the number of accepted ordered pairs producing z is

$$\#\{(s, t) \in [N_{i-1}]^2 : x_s - x_t = z, h_i(x_s - x_t) = 1\}.$$

Hence the empirical mass of z in L_i is

$$\mu_i(z) = \frac{\#\{(s, t) : x_s - x_t = z, h_i(x_s - x_t) = 1\}}{\#\{(s, t) : h_i(x_s - x_t) = 1\}}.$$

On the other hand, sampling X, Y independently from μ_{i-1} is exactly the same as choosing an ordered pair $(s, t) \in [N_{i-1}]^2$ uniformly at random and outputting x_s, x_t . Therefore

$$\Pr[X - Y = z, h_i(X - Y) = 1] = \frac{1}{N_{i-1}^2} \#\{(s, t) : x_s - x_t = z, h_i(x_s - x_t) = 1\},$$

and

$$\Pr[h_i(X - Y) = 1] = \frac{1}{N_{i-1}^2} \#\{(s, t) : h_i(x_s - x_t) = 1\}.$$

Dividing the two displays gives the claimed formula for $\mu_i(z)$.

It remains to note that $\alpha_i > 0$. Since L_{i-1} is nonempty and *self*-pairs (s, s) satisfy $x_s - x_s = 0$, and since $h_i(0) = 1$, at least one ordered pair is accepted. $\square$

Corollary 3.8 (Mass at zero and finite-list success). *Let $N_i := |L_i|$, counted with multiplicity. Then the number of zero entries in L_i is exactly $N_i \mu_i(0)$, and the number of nonzero entries is exactly $N_i(1 - \mu_i(0))$. Consequently, if*

$$\mu_i(0) \leq \frac{N_i - 1}{N_i},$$

then L_i contains at least one nonzero vector. In particular, if $N_i \geq 2$ and $\mu_i(0) \leq 1/2$, then L_i contains a nonzero vector.

Proof. By definition of the empirical law,

$$N_i \mu_i(0) = \#\{s \in [N_i] : X_s^{(i)} = 0\}.$$

Subtracting from N_i gives

$$N_i(1 - \mu_i(0)) = \#\{s \in [N_i] : X_s^{(i)} \neq 0\}.$$

If $\mu_i(0) \leq (N_i - 1)/N_i$, then $N_i(1 - \mu_i(0)) \geq 1$, so at least one list entry is nonzero. The final assertion follows because $1/2 \leq (N_i - 1)/N_i$ whenever $N_i \geq 2$. $\square$

Remark 3.9 (Population and empirical level zero). There are two level-zero distributions in the paper.

First, the population ternary law

$$\mu_0^{\text{pop}} := \text{Unif}(\{-1, 0, 1\}^d)$$

is useful for clean deterministic Fourier computations.

Second, the actual finite algorithm starts from the empirical law

$$\mu_0 := \mu_{L_0}.$$

The exact list-distribution identity in Lemma 3.7 applies to this empirical μ_0 , not directly to μ_0^{pop} .

The empirical distribution $\mu_0 = \mu_{L_0}$ is generally not even, because the sample counts of x and $-x$ need not agree. However, for any distribution μ , the difference law $\mu * \check{\mu}$ is even. Indeed,

$$(\mu * \check{\mu})(x) = \sum_{u \in G} \mu(u) \mu(u - x)$$

and, after the substitution $v = u + x$,

$$(\mu * \check{\mu})(-x) = \sum_{u \in G} \mu(u) \mu(u + x) = \sum_{v \in G} \mu(v - x) \mu(v) = (\mu * \check{\mu})(x).$$

Since each cube C_i is centrally symmetric, the filter $h_i(x) = \mathbf{1}_{C_i}(-A'_i x)$ satisfies $h_i(-x) = h_i(x)$. Therefore conditioning preserves evenness. Hence μ_i is even for every $i \geq 1$, although the empirical μ_0 need not be even.

3.3 Parameter regime

We use the following asymptotic regime, parameterized by n .

Definition 3.10 (Parameter regime). Fix constants $\kappa > 0$ and $0 < c_f \leq C_f < \infty$. Let $q = q(n)$ be an odd prime satisfying

$$q = n^{\kappa + o(1)}.$$

Let $f = f(n)$ satisfy

$$(\log n)^{c_f} \leq f(n) \leq (\log n)^{C_f}.$$

Set

$$\beta := \frac{q}{f}, \quad r := \lfloor \log_2(q/f) \rfloor - 1, \quad d := m - n = \omega\left(\frac{n}{\log \log n}\right).$$

For level i , define

$$B_i := 2^i, \quad W_i := 2B_i + 1 = 2^{i+1} + 1.$$

Thus $C_i = \{y \in \mathbb{Z}_q^{b_i} : \|y\|_{\infty, q} \leq B_i\}$ has size

$$|C_i| = W_i^{b_i}.$$

The block sizes are chosen according to the survival probability of the simplified full-pair filter. Under a uniform-syndrome heuristic, a random difference survives level i with probability

$$\frac{|C_i|}{q^{b_i}} = \left(\frac{W_i}{q}\right)^{b_i}.$$

Since the virtual list has roughly N^2 ordered pairs, keeping the list size near N suggests

$$N^2 \left(\frac{W_i}{q}\right)^{b_i} \approx N,$$

or equivalently

$$N \approx \left(\frac{q}{W_i}\right)^{b_i}.$$

Proposition 3.11 (Balanced block sizes). *Ignoring integer rounding, define*

$$b_i := \frac{\log_2 N}{\log_2(q/W_i)}, \quad i = 1, \dots, r.$$

The constraint $\sum_{i=1}^r b_i = n$ is equivalent to

$$\log_2 N = \frac{n}{\sum_{i=1}^r \frac{1}{\log_2(q/W_i)}}.$$

Moreover, under Definition 3.10,

$$\sum_{i=1}^r \frac{1}{\log_2(q/W_i)} = \ln \ln q - \ln \ln f + O(1),$$

and hence

$$\log_2 N = \frac{n}{\ln \ln q - \ln \ln f} (1 + o(1)) = \Theta\left(\frac{n}{\log \log n}\right).$$

Proof. The equivalence between $\sum_i b_i = n$ and the displayed expression for $\log_2 N$ follows by substituting $b_i = \log_2 N / \log_2(q/W_i)$ and solving for $\log_2 N$.

It remains to estimate the sum. Write $L := \log_2 q$. Since

$$W_i = 2^{i+1} + 1 = 2^{i+1}(1 + 2^{-i-1}),$$

we have

$$\log_2(q/W_i) = L - i - 1 - \log_2(1 + 2^{-i-1}).$$

The final term is bounded uniformly by an absolute constant. Hence the sum is estimated by the same integral as

$$\sum_{i=1}^r \frac{1}{L - i + O(1)}.$$

By integral comparison for the increasing function $x \mapsto (L - x)^{-1}$ on the relevant range,

$$\sum_{i=1}^r \frac{1}{L - i + O(1)} = \ln\left(\frac{L + O(1)}{L - r + O(1)}\right) + O(1).$$

Because

$$r = \lfloor \log_2(q/f) \rfloor - 1,$$

we have

$$L - r + O(1) = \log_2 f + O(1).$$

Therefore

$$\sum_{i=1}^r \frac{1}{\log_2(q/W_i)} = \ln(\log_2 q) - \ln(\log_2 f) + O(1) = \ln \ln q - \ln \ln f + O(1),$$

since the additive constants coming from $\log_2 t = (\ln t)/(\ln 2)$ cancel.

Finally, for $q = n^{\kappa+o(1)}$ and $(\log n)^{c_f} \leq f \leq (\log n)^{C_f}$,

$$\ln \ln q - \ln \ln f = \Theta(\log \log n),$$

which gives the claimed order of $\log_2 N$. □

Remark 3.12 (Integer block sizes). The formulas above define real-valued block sizes. In the actual algorithm, choose integer block sizes by rounding $b_1, \dots, b_{r-1}$ and setting $b_r := n - \sum_{i=1}^{r-1} b_i$. Since $r = O(\log n)$ and each displayed estimate below is stable under additive $O(1)$ perturbations of the b_i , this integerization does not affect any asymptotic bound used in the paper.

Lemma 3.13 (Block-size asymptotics). *Under Definition 3.10, the following hold for every $i \in \{1, \dots, r\}$:*

(i) $b_i = O(n/(\log \log n)^2)$. In particular, $b_i = o(d)$.

(ii)

$$b_i \log(q/W_i) = \Theta\left(\frac{n}{\log \log n}\right).$$

Consequently,

$$b_i \log(q/2^i) = \Theta\left(\frac{n}{\log \log n}\right).$$

(iii) $W_i \leq q/8$ for all sufficiently large n .

Proof. By Proposition 3.11,

$$\log_2 N = \Theta\left(\frac{n}{\log \log n}\right).$$

For $i \leq r$,

$$W_i = 2^{i+1} + 1 \leq 2^{r+1} + 1 \leq q/f + 1.$$

Since $f \rightarrow \infty$, for all sufficiently large n this gives

$$\log_2(q/W_i) = \Omega(\log_2 f) = \Omega(\log \log n).$$

Thus

$$b_i = \frac{\log_2 N}{\log_2(q/W_i)} = O\left(\frac{n}{(\log \log n)^2}\right).$$

Because $d = \omega(n/\log \log n)$, this implies $b_i = o(d)$.

For the second claim, the definition of b_i gives exactly

$$b_i \log(q/W_i) = \log N = \Theta\left(\frac{n}{\log \log n}\right).$$

Also,

$$W_i = 2^i(2 + 2^{-i}),$$

so

$$\log(q/W_i) = \log(q/2^i) + O(1).$$

Multiplying this $O(1)$ discrepancy by $b_i = O(n/(\log \log n)^2)$ is lower order than $n/\log \log n$, proving the corresponding estimate for $b_i \log(q/2^i)$.

For the third claim, the bound $W_i \leq q/f + 1$ and $f \rightarrow \infty$ imply $W_i \leq q/8$ for all sufficiently large n . $\square$

Lemma 3.14 (Admissibility-scale consequences). *Under Definition 3.10, for every $i \in \{1, \dots, r\}$,*

$$b_i \log\left(\frac{q}{W_i}\right) = \Theta\left(\frac{n}{\log \log n}\right),$$

and

$$b_i \log\left(\frac{C_{\ker} q \log W_i}{W_i}\right) = \Theta\left(\frac{n}{\log \log n}\right).$$

Proof. The first estimate is Lemma 3.13(ii).

For the second, decompose

$$\log\left(\frac{C_{\ker} q \log W_i}{W_i}\right) = \log(q/W_i) + \log C_{\ker} + \log \log W_i.$$

Multiplication by b_i gives three contributions. The first is $\Theta(n/\log \log n)$ by the previous paragraph. The second is $O(b_i) = O(n/(\log \log n)^2)$ by Lemma 3.13(i). For the third, since $W_i \leq q$ and $q = n^{\kappa+o(1)}$,

$$\log \log W_i = O(\log \log q) = O(\log \log n).$$

Therefore

$$b_i \log \log W_i = O\left(\frac{n}{(\log \log n)^2}\right) O(\log \log n) = O\left(\frac{n}{\log \log n}\right).$$

The lower bound follows from the positive first contribution $b_i \log(q/W_i) = \Theta(n/\log \log n)$, so the whole expression is $\Theta(n/\log \log n)$.

Together with $b_i = o(d)$ and $W_i \leq q/8$ from Lemma 3.13, this proves all admissibility clauses. $\square$

4 Spectral Analysis of the Initial Distribution

This section first studies the population ternary law

$$\mu_0^{\text{pop}} := \text{Unif}(\{-1, 0, 1\}^d).$$

For notational economy, in Sections 4.1 and 4.2 and in the population tail bounds we write μ_0 for μ_0^{pop} . The actual finite algorithm uses the empirical law μ_{L_0} ; the empirical bounds in Lemmas 4.8 and 4.11 are the corresponding level-zero inputs for the finite-list analysis. This section computes the Fourier transform $\widehat{\mu_0}$ exactly, determines its extremal magnitudes, and establishes tail bounds on the spectral density $\delta_0(\theta)$.

4.1 Fourier transform of μ_0

Definition 4.1. Define $g : \mathbb{Z}_q \rightarrow \mathbb{R}$ by

$$g(y) := \frac{1 + 2 \cos(2\pi y/q)}{3}.$$

Proposition 4.2. *For every $\omega = (\omega_1, \dots, \omega_d) \in \mathbb{Z}_q^d$,*

$$\widehat{\mu_0}(\omega) = \prod_{j=1}^d g(\omega_j). \tag{5}$$

In particular, $\widehat{\mu_0}(\omega) \in \mathbb{R}$ and $\widehat{\mu_0}$ is even.

Proof. Since the coordinates $X_1, \dots, X_d$ of $X \sim \mu_0$ are independent and each uniform on $\{-1, 0, 1\}$,

$$\widehat{\mu_0}(\omega) = \prod_{j=1}^d \mathbb{E} \left[e^{2\pi i \omega_j X_j / q} \right].$$

For a single coordinate,

$$\mathbb{E} \left[e^{2\pi i \omega_j X_j / q} \right] = \frac{1}{3} (e^{-2\pi i \omega_j / q} + 1 + e^{2\pi i \omega_j / q}) = \frac{1 + 2 \cos(2\pi \omega_j / q)}{3} = g(\omega_j).$$

Each factor is real, so $\widehat{\mu_0}(\omega) \in \mathbb{R}$. Evenness follows from $g(-y) = g(y)$. $\square$

4.2 Extremal magnitudes

Proposition 4.3 (Worst-case magnitude).

$$\max_{\omega \neq 0} |\widehat{\mu_0}(\omega)| = g(1) = \frac{1 + 2 \cos(2\pi/q)}{3} = 1 - \frac{4\pi^2}{3q^2} + O(q^{-4}).$$

The maximum is attained by any weight-1 frequency (e.g. $\omega = e_1$).

Proof. From (5), $|\widehat{\mu_0}(\omega)| = \prod_j |g(\omega_j)|$. Since $g(0) = 1$ and $|g(y)| < 1$ for every nonzero $y \in \mathbb{Z}_q$, the product $\prod_j |g(\omega_j)|$ is maximized over $\omega \neq 0$ by taking exactly one nonzero coordinate and all others equal to 0. Thus

$$\max_{\omega \neq 0} |\widehat{\mu_0}(\omega)| = \max_{y \in \mathbb{Z}_q \setminus \{0\}} |g(y)|.$$

Now

$$g(y) = \frac{1 + 2 \cos(2\pi y/q)}{3} \in [-1/3, 1].$$

Hence the largest absolute value away from 0 must occur on the positive branch, where $g(y) \geq 0$ and $\cos(2\pi y/q)$ is largest. By symmetry $g(-y) = g(y)$, and among nonzero residues the maximum of $g(y)$ is attained at $y = \pm 1$. The Taylor expansion $\cos x = 1 - x^2/2 + O(x^4)$ with $x = 2\pi/q$ gives $g(1) = 1 - 4\pi^2/(3q^2) + O(q^{-4})$. $\square$

Remark 4.4. Since $q = \text{poly}(n)$, the worst-case magnitude $g(1) = 1 - \Theta(q^{-2})$ is asymptotically 1. Any proof strategy requiring the global bias $\max_{\omega \neq 0} |\widehat{\mu_0}(\omega)|$ to be uniformly small is therefore fatally obstructed at level 0.

Proposition 4.5 (Small Fourier coefficients). *Assume $3 \nmid q$, and let $y_* := \lfloor q/3 \rfloor$. Then*

$$|g(y_*)| = \frac{2\pi}{3\sqrt{3}q} + O(q^{-2}).$$

Consequently,

$$\min_{\omega \in \mathbb{Z}_q^d} |\widehat{\mu_0}(\omega)| \leq |g(y_*)|^d = \exp(-d \log q + O(d)).$$

The displayed upper bound is obtained by taking $\omega = (y_*, \dots, y_*)$.

Proof. Write $y_* = q/3 + \delta$ with $|\delta| \leq 1/2$. Then $2\pi y_*/q = 2\pi/3 + 2\pi\delta/q$, so

$$\cos(2\pi y_*/q) = \cos(2\pi/3) \cos(2\pi\delta/q) - \sin(2\pi/3) \sin(2\pi\delta/q) = -\frac{1}{2} + O(q^{-2}) - \frac{\sqrt{3}}{2} \cdot \frac{2\pi\delta}{q} + O(q^{-2}).$$

Substituting into $g(y_*) = (1 + 2\cos(2\pi y_*/q))/3$ gives $g(y_*) = -2\pi\delta\sqrt{3}/(3q) + O(q^{-2})$. Taking $\omega = (y_*, \dots, y_*)$ in

$$\widehat{\mu}_0(\omega) = \prod_{j=1}^d g(\omega_j)$$

therefore gives

$$\min_{\omega \in \mathbb{Z}_q^d} |\widehat{\mu}_0(\omega)| \leq |g(y_*)|^d = \exp(-d \log q + O(d)).$$

□

4.3 Tail bounds on the spectral density

The spectral density $\delta_0(\theta)$ measures the fraction of frequencies exceeding a given magnitude threshold.

Lemma 4.6. $\mathbb{E}[|g(\omega)|^2] = 1/3$ for ω uniform on $\mathbb{Z}_q$ (for $q \geq 5$).

Proof. By direct computation,

$$\mathbb{E}[|g(\omega)|^2] = \frac{1}{q} \sum_{y=0}^{q-1} \left(\frac{1 + 2\cos(2\pi y/q)}{3} \right)^2 = \frac{1}{9q} \sum_{y=0}^{q-1} (1 + 4\cos(2\pi y/q) + 4\cos^2(2\pi y/q)).$$

Using $\sum_y \cos(2\pi y/q) = 0$ and $\sum_y \cos^2(2\pi y/q) = q/2$ (valid for $q \geq 3$), the sum evaluates to $q(1 + 0 + 2)/9 = q/3$, giving $\mathbb{E}[|g(\omega)|^2] = 1/3$. □

Theorem 4.7 (L^2 spectral tail bound). *For all $\theta \in (0, 1)$ and $q \geq 5$,*

$$\delta_0(\theta) \leq \frac{1}{3^d \theta^2}. \quad (6)$$

Proof. By Markov's inequality applied to $|\widehat{\mu}_0(\omega)|^2$ under the uniform measure on G ,

$$\delta_0(\theta) = \Pr_{\omega} [|\widehat{\mu}_0(\omega)| > \theta] \leq \frac{\mathbb{E}_{\omega} [|\widehat{\mu}_0(\omega)|^2]}{\theta^2}.$$

The expectation factors by independence of coordinates:

$$\mathbb{E}_{\omega} [|\widehat{\mu}_0(\omega)|^2] = \prod_{j=1}^d \mathbb{E}_{\omega_j} [g(\omega_j)^2] = (1/3)^d,$$

using Lemma 4.6. This gives (6). □

Lemma 4.8 (Empirical level-zero spectral tail). *Let $q \geq 5$, $N_0 \geq 1$, and $\eta > 1$. Let $X_1, \dots, X_{N_0}$ be independent samples from $\mu_0 = \text{Unif}(\{-1, 0, 1\}^d)$. Define the empirical level-zero distribution μ_0^{emp} by*

$$\mu_0^{\text{emp}}(x) := \frac{1}{N_0} \#\{t \in \{1, \dots, N_0\} : X_t = x\}, \quad x \in G.$$

Equivalently, sampling from μ_0^{emp} means choosing $t \in \{1, \dots, N_0\}$ uniformly at random and outputting X_t .

For $\theta \in (0, 1)$, define

$$\delta_0^{\text{emp}}(\theta) := \frac{1}{q^d} \# \{ \omega \in \mathbb{Z}_q^d : |\widehat{\mu_0^{\text{emp}}}(\omega)| > \theta \}.$$

Then, with probability at least $1 - 1/\eta$ over the sampled list $X_1, \dots, X_{N_0}$, one has simultaneously for all $\theta \in (0, 1)$,

$$\delta_0^{\text{emp}}(\theta) \leq \frac{\eta}{\theta^2} \left(\frac{1}{N_0} + \left(1 - \frac{1}{N_0}\right) 3^{-d} \right) \leq \frac{\eta}{\theta^2} \left(\frac{1}{N_0} + 3^{-d} \right). \quad (7)$$

Proof. For each $\omega \in \mathbb{Z}_q^d$, the Fourier coefficient of the empirical distribution is

$$\widehat{\mu_0^{\text{emp}}}(\omega) = \frac{1}{N_0} \sum_{t=1}^{N_0} \chi_\omega(X_t).$$

For a fixed ω , write

$$Y_t(\omega) := \chi_\omega(X_t).$$

Then $|Y_t(\omega)| = 1$ and $\mathbb{E}[Y_t(\omega)] = \widehat{\mu_0}(\omega)$. Therefore

$$\begin{aligned} \mathbb{E}[|\widehat{\mu_0^{\text{emp}}}(\omega)|^2] &= \mathbb{E}\left[\left|\frac{1}{N_0} \sum_{t=1}^{N_0} Y_t(\omega)\right|^2\right] \\ &= \frac{1}{N_0^2} \sum_{s,t=1}^{N_0} \mathbb{E}[Y_s(\omega) \overline{Y_t(\omega)}]. \end{aligned}$$

The diagonal terms $s = t$ contribute 1, since $|Y_t(\omega)| = 1$. For $s \neq t$, independence gives

$$\mathbb{E}[Y_s(\omega) \overline{Y_t(\omega)}] = \mathbb{E}[Y_s(\omega)] \overline{\mathbb{E}[Y_t(\omega)]} = |\widehat{\mu_0}(\omega)|^2.$$

Hence

$$\mathbb{E}[|\widehat{\mu_0^{\text{emp}}}(\omega)|^2] = \frac{1}{N_0} + \left(1 - \frac{1}{N_0}\right) |\widehat{\mu_0}(\omega)|^2.$$

Averaging over $\omega \in \mathbb{Z}_q^d$ gives

$$\begin{aligned} \mathbb{E}\left[\frac{1}{q^d} \sum_{\omega \in \mathbb{Z}_q^d} |\widehat{\mu_0^{\text{emp}}}(\omega)|^2\right] &= \frac{1}{N_0} + \left(1 - \frac{1}{N_0}\right) \frac{1}{q^d} \sum_{\omega \in \mathbb{Z}_q^d} |\widehat{\mu_0}(\omega)|^2 \\ &= \frac{1}{N_0} + \left(1 - \frac{1}{N_0}\right) 3^{-d}, \end{aligned}$$

where the last equality follows from Proposition 4.2 and Lemma 4.6.

By Markov's inequality, with probability at least $1 - 1/\eta$,

$$\frac{1}{q^d} \sum_{\omega \in \mathbb{Z}_q^d} |\widehat{\mu_0^{\text{emp}}}(\omega)|^2 \leq \eta \left(\frac{1}{N_0} + \left(1 - \frac{1}{N_0}\right) 3^{-d} \right).$$

On this event, for every $\theta \in (0, 1)$,

$$\theta^2 \delta_0^{\text{emp}}(\theta) \leq \frac{1}{q^d} \sum_{\omega \in \mathbb{Z}_q^d} |\widehat{\mu_0^{\text{emp}}}(\omega)|^2.$$

This proves (7). □

We use the notation $[z^s]F(z)$ for the coefficient of z^s in a Laurent polynomial $F(z)$. The variable z below is only a formal bookkeeping variable; it will later be specialized to $z = e^{2\pi i y/q}$.

Lemma 4.9 (Higher even moments of the level-zero spectrum). *Let $\ell \geq 1$ and set $p := 2^\ell$. Define*

$$M_p(q) := \mathbb{E}_{y \sim \text{Unif}(\mathbb{Z}_q)} [|g(y)|^p].$$

Then

$$\mathbb{E}_{\omega \sim \text{Unif}(G)} [|\widehat{\mu_0}(\omega)|^p] = M_p(q)^d. \quad (8)$$

Moreover, let u be a formal Laurent-polynomial variable, and write $[u^s]F(u)$ for the coefficient of u^s in a Laurent polynomial $F(u)$. Then

$$M_p(q) = 3^{-p} \sum_{\substack{s \in \mathbb{Z} \\ |s| \leq p \\ q|s}} [u^s](1 + u + u^{-1})^p. \quad (9)$$

In particular, if $p < q$, then

$$M_p(q) = 3^{-p} [u^0](1 + u + u^{-1})^p = 3^{-p} \sum_{a=0}^{\lfloor p/2 \rfloor} \frac{p!}{a! a! (p-2a)!}. \quad (10)$$

Proof. By Proposition 4.2,

$$\widehat{\mu_0}(\omega) = \prod_{j=1}^d g(\omega_j).$$

Since ω is uniform on $G = \mathbb{Z}_q^d$, its coordinates are independent and uniform on $\mathbb{Z}_q$. Therefore

$$\begin{aligned} \mathbb{E}_{\omega \sim \text{Unif}(G)} [|\widehat{\mu_0}(\omega)|^p] &= \mathbb{E}_{\omega} \left[\prod_{j=1}^d |g(\omega_j)|^p \right] \\ &= \prod_{j=1}^d \mathbb{E}_{\omega_j \sim \text{Unif}(\mathbb{Z}_q)} [|g(\omega_j)|^p] \\ &= M_p(q)^d. \end{aligned}$$

This proves (8).

It remains to compute $M_p(q)$. Since $p = 2^\ell$ and $\ell \geq 1$, p is even, so $|g(y)|^p = g(y)^p$. Let

$$\zeta := e^{2\pi i/q}.$$

Since

$$1 + 2 \cos(2\pi y/q) = 1 + \zeta^y + \zeta^{-y},$$

we have

$$M_p(q) = 3^{-p} \frac{1}{q} \sum_{y \in \mathbb{Z}_q} (1 + \zeta^y + \zeta^{-y})^p.$$

Now introduce the formal Laurent polynomial

$$(1 + u + u^{-1})^p = \sum_{s=-p}^p [u^s](1 + u + u^{-1})^p u^s.$$

Substituting $u = \zeta^y$ gives

$$(1 + \zeta^y + \zeta^{-y})^p = \sum_{s=-p}^p [u^s](1 + u + u^{-1})^p \zeta^{sy}.$$

Averaging over $y \in \mathbb{Z}_q$ and using character orthogonality,

$$\frac{1}{q} \sum_{y \in \mathbb{Z}_q} \zeta^{sy} = \begin{cases} 1, & q \mid s, \\ 0, & q \nmid s, \end{cases}$$

we obtain

$$M_p(q) = 3^{-p} \sum_{\substack{s \in \mathbb{Z} \\ |s| \leq p \\ q \mid s}} [u^s](1 + u + u^{-1})^p.$$

This proves (9).

If $p < q$, then every integer s satisfying $|s| \leq p$ and $q \mid s$ must be $s = 0$. Hence only the zero exponent survives in (9), and

$$M_p(q) = 3^{-p} [u^0](1 + u + u^{-1})^p.$$

It remains only to compute this coefficient. To obtain total exponent zero, choose a copies of u , a copies of u^{-1} , and $p - 2a$ copies of 1. Summing over all possible a gives

$$[u^0](1 + u + u^{-1})^p = \sum_{a=0}^{\lfloor p/2 \rfloor} \frac{p!}{a! a! (p - 2a)!}.$$

This proves (10). □

Theorem 4.10 (Higher-moment spectral tail bound). *Let $\ell \geq 1$, set $p := 2^\ell$, and let $\theta \in (0, 1)$. No relation between p and q is assumed. Then*

$$\delta_0(\theta) \leq \frac{M_p(q)^d}{\theta^p}, \tag{11}$$

where $M_p(q)$ is defined in Lemma 4.9. If in addition $p < q$, then

$$\delta_0(\theta) \leq \theta^{-2^\ell} \left(3^{-2^\ell} [u^0](1 + u + u^{-1})^{2^\ell} \right)^d. \tag{12}$$

Equivalently, in the range $2^\ell < q$,

$$\delta_0(\theta) \leq \theta^{-2^\ell} \left(3^{-2^\ell} \sum_{a=0}^{2^{\ell-1}} \frac{(2^\ell)!}{a! a! (2^\ell - 2a)!} \right)^d. \tag{13}$$

Proof. By Markov's inequality applied to $|\widehat{\mu_0}(\omega)|^p$ under the uniform measure on G ,

$$\delta_0(\theta) = \Pr_\omega [|\widehat{\mu_0}(\omega)| > \theta] = \Pr_\omega [|\widehat{\mu_0}(\omega)|^p > \theta^p] \leq \frac{\mathbb{E}_\omega [|\widehat{\mu_0}(\omega)|^p]}{\theta^p}.$$

Using Lemma 4.9,

$$\mathbb{E}_\omega [|\widehat{\mu_0}(\omega)|^p] = M_p(q)^d.$$

This proves (11). The clean forms (12) and (13) follow from (10). □

Lemma 4.11 (Empirical higher-moment level-zero spectral tail). *Let $q \geq 5$, $N_0 \geq 1$, and $\eta > 1$. Let $X_1, \dots, X_{N_0}$ be independent samples from $\mu_0 = \text{Unif}(\{-1, 0, 1\}^d)$, and let μ_0^{emp} be the corresponding empirical distribution. Let $\ell \geq 1$ and set $p := 2^\ell$. Set*

$$C_{\text{emp}} := 2\sqrt{2}.$$

Then, with probability at least $1 - 1/\eta$ over the sampled list $X_1, \dots, X_{N_0}$, one has simultaneously for all $\theta \in (0, 1)$,

$$\delta_0^{\text{emp}}(\theta) \leq \frac{\eta}{\theta^p} \left(M_p(q)^{d/p} + C_{\text{emp}} \sqrt{\frac{p}{N_0}} \right)^p, \quad (14)$$

where $M_p(q)$ is defined in Lemma 4.9. In particular, if $p < q$, then $M_p(q)$ is given by the central trinomial expression in (10).

Proof. For each $\omega \in G$, write

$$\widehat{\mu_0^{\text{emp}}}(\omega) = \frac{1}{N_0} \sum_{t=1}^{N_0} \chi_\omega(X_t).$$

Add and subtract the population Fourier coefficient:

$$\widehat{\mu_0^{\text{emp}}}(\omega) = \widehat{\mu_0}(\omega) + R(\omega),$$

where

$$R(\omega) := \frac{1}{N_0} \sum_{t=1}^{N_0} (\chi_\omega(X_t) - \widehat{\mu_0}(\omega)).$$

For fixed ω , the summands in $R(\omega)$ are independent, mean-zero, complex-valued random variables. Moreover, $\chi_\omega(X_t)$ has modulus 1, so both its real and imaginary parts lie in $[-1, 1]$.

We first record a standard bounded-moment estimate for $R(\omega)$. By Hoeffding's inequality applied separately to the real and imaginary parts, for every $u > 0$,

$$\Pr[|R(\omega)| \geq u] \leq 4 \exp(-N_0 u^2/4).$$

Indeed, if $|R(\omega)| \geq u$, then either $|\Re R(\omega)| \geq u/\sqrt{2}$ or $|\Im R(\omega)| \geq u/\sqrt{2}$, and Hoeffding's inequality gives the displayed bound for each part.

Using the tail-integral formula for nonnegative random variables,

$$\mathbb{E}[|R(\omega)|^p] = p \int_0^\infty u^{p-1} \Pr[|R(\omega)| \geq u] du,$$

we obtain

$$\mathbb{E}[|R(\omega)|^p] \leq 4p \int_0^\infty u^{p-1} \exp(-N_0 u^2/4) du$$

Using the substitution $x = N_0 u^2/4$, and using that $p/2$ is a positive integer, we obtain

$$\begin{aligned} \int_0^\infty u^{p-1} \exp(-N_0 u^2/4) du &= \frac{1}{2} \left(\frac{4}{N_0} \right)^{p/2} \int_0^\infty x^{p/2-1} e^{-x} dx \\ &= \frac{1}{2} \left(\frac{4}{N_0} \right)^{p/2} \left(\frac{p}{2} - 1 \right)!. \end{aligned}$$

Therefore

$$\begin{aligned}\mathbb{E}[|R(\omega)|^p] &\leq 4p \int_0^\infty u^{p-1} \exp(-N_0 u^2/4) du \\ &= 2p \left(\frac{4}{N_0}\right)^{p/2} \left(\frac{p}{2} - 1\right)!. \end{aligned}$$

Since $p/2$ is a positive integer,

$$\left(\frac{p}{2} - 1\right)! \leq \left(\frac{p}{2}\right)^{p/2}.$$

Therefore

$$\begin{aligned}(\mathbb{E}[|R(\omega)|^p])^{1/p} &\leq \left(2p \left(\frac{4}{N_0}\right)^{p/2} \left(\frac{p}{2}\right)^{p/2}\right)^{1/p} \\ &= (2p)^{1/p} \sqrt{\frac{2p}{N_0}}. \end{aligned}$$

For $p \geq 2$, the function $p \mapsto (2p)^{1/p}$ is decreasing, and therefore

$$(2p)^{1/p} \leq 4^{1/2} = 2.$$

Hence

$$(\mathbb{E}[|R(\omega)|^p])^{1/p} \leq 2\sqrt{2} \sqrt{\frac{p}{N_0}} = C_{\text{emp}} \sqrt{\frac{p}{N_0}}. \quad (15)$$

Now average also over $\omega \sim \text{Unif}(G)$ and use Minkowski's inequality in L^p :

$$\begin{aligned}\left(\mathbb{E}_{X_1, \dots, X_{N_0}} \mathbb{E}_{\omega \sim \text{Unif}(G)} \left[|\widehat{\mu}_0^{\text{emp}}(\omega)|^p\right]\right)^{1/p} &= \left(\mathbb{E}_{X_1, \dots, X_{N_0}, \omega} [|\widehat{\mu}_0(\omega) + R(\omega)|^p]\right)^{1/p} \\ &\leq (\mathbb{E}_\omega [|\widehat{\mu}_0(\omega)|^p])^{1/p} + \left(\mathbb{E}_{X_1, \dots, X_{N_0}, \omega} [|R(\omega)|^p]\right)^{1/p}. \end{aligned}$$

By Lemma 4.9,

$$\mathbb{E}_\omega [|\widehat{\mu}_0(\omega)|^p] = M_p(q)^d.$$

Together with (15), this gives

$$\mathbb{E}_{X_1, \dots, X_{N_0}} \left[\frac{1}{q^d} \sum_{\omega \in G} |\widehat{\mu}_0^{\text{emp}}(\omega)|^p \right] \leq \left(M_p(q)^{d/p} + C_{\text{emp}} \sqrt{\frac{p}{N_0}} \right)^p.$$

By Markov's inequality, with probability at least $1 - 1/\eta$,

$$\frac{1}{q^d} \sum_{\omega \in G} |\widehat{\mu}_0^{\text{emp}}(\omega)|^p \leq \eta \left(M_p(q)^{d/p} + C_{\text{emp}} \sqrt{\frac{p}{N_0}} \right)^p.$$

On this event, for every $\theta \in (0, 1)$,

$$\theta^p \delta_0^{\text{emp}}(\theta) \leq \frac{1}{q^d} \sum_{\omega \in G} |\widehat{\mu}_0^{\text{emp}}(\omega)|^p.$$

Combining the last two displays proves (14). □

5 The Fourier Update Rule

Fix a level $i \in \{1, \dots, r\}$. Write

$$G := \mathbb{Z}_q^d, \quad H := \mathbb{Z}_q^{b_i}, \quad M := A'_i \in \mathbb{Z}_q^{b_i \times d},$$

and let $B := 2^i$, $W := 2B + 1 = W_i$, and $C := C_i = \{y \in H : \|y\|_{\infty, q} \leq B\}$. Let $\mu := \mu_{i-1}$ denote the level- $(i-1)$ distribution on G .

Each round of the algorithm performs two operations on μ : differencing (forming the virtual list) and conditioning (imposing the norm constraint). We compute the Fourier-domain effect of each operation exactly.

5.1 Fourier transform of the conditioning operator

Definition 5.1. The *conditioning operator* at level i is

$$h_i : G \rightarrow \{0, 1\}, \quad h_i(x) := \mathbf{1}_C(-Mx).$$

Lemma 5.2 (Fourier expansion of h_i).

$$h_i(x) = \frac{1}{q^{b_i}} \sum_{\tau \in H} \widehat{\mathbf{1}_C}(\tau) \chi_{M^T \tau}(x), \quad (16)$$

where $\widehat{\mathbf{1}_C}$ is the unnormalized Fourier transform on H , given by

$$\widehat{\mathbf{1}_C}(\tau) = \prod_{j=1}^{b_i} D_B(\tau_j), \quad \tau = (\tau_1, \dots, \tau_{b_i}) \in H. \quad (17)$$

Proof. By Fourier inversion on H (Equation (2)),

$$\mathbf{1}_C(y) = \frac{1}{q^{b_i}} \sum_{\tau \in H} \widehat{\mathbf{1}_C}(\tau) \chi_{-\tau}(y).$$

Substitute $y = -Mx$:

$$h_i(x) = \frac{1}{q^{b_i}} \sum_{\tau \in H} \widehat{\mathbf{1}_C}(\tau) \chi_{-\tau}(-Mx) = \frac{1}{q^{b_i}} \sum_{\tau \in H} \widehat{\mathbf{1}_C}(\tau) \chi_{M^T \tau}(x),$$

using $\langle \tau, Mx \rangle = \langle M^T \tau, x \rangle$. For (17), since $C = \{y \in H : \|y\|_{\infty, q} \leq B\}$ factors coordinate-wise as $[-B, B]^{b_i}$ in centered representatives (valid when $B < q/2$, which holds by Lemma 3.13(iii)),

$$\widehat{\mathbf{1}_C}(\tau) = \sum_{y \in C} \chi_{\tau}(y) = \prod_{j=1}^{b_i} \sum_{s=-B}^B e^{2\pi i \tau_j s/q} = \prod_{j=1}^{b_i} D_B(\tau_j). \quad \square$$

Definition 5.3. The *level- i kernel* is $K_i(\tau) := \widehat{\mathbf{1}_C}(\tau) = \prod_{j=1}^{b_i} D_B(\tau_j)$.

Proposition 5.4 (Fourier transform of h_i on G). *The unnormalized Fourier transform of h_i on G is*

$$\widehat{h_i}(\omega) = q^{d-b_i} \sum_{\substack{\tau \in H \\ M^T \tau = \omega}} K_i(\tau). \quad (18)$$

In particular:

(i) $\widehat{h}_i(\omega) = 0$ unless $\omega \in \text{im}(M^T)$.

(ii) If M has full row rank b_i (equivalently, M^T is injective), then for each $\omega \in \text{im}(M^T)$ there is a unique $\tau \in H$ with $M^T \tau = \omega$, and

$$\widehat{h}_i(\omega) = q^{d-b_i} K_i(\tau). \quad (19)$$

Proof. By definition, $\widehat{h}_i(\omega) = \sum_{x \in G} h_i(x) \chi_\omega(x)$. Substitute (16):

$$\widehat{h}_i(\omega) = \frac{1}{q^{b_i}} \sum_{\tau \in H} K_i(\tau) \underbrace{\sum_{x \in G} \chi_{\omega + M^T \tau}(x)}_{= q^d \mathbf{1}[\omega + M^T \tau = 0]},$$

by character orthogonality (1). Hence only terms with $M^T \tau = -\omega$ survive. Using the evenness $K_i(-\tau) = K_i(\tau)$ (Lemma 2.10(iii)), the substitution $\tau \mapsto -\tau$ converts the condition to $M^T \tau = \omega$, giving (18). Parts (i) and (ii) are immediate. $\square$

Remark 5.5 (Fourier support of the filter). Proposition 5.4 identifies the Fourier support of the physical-space filter h_i . Namely, $\widehat{h}_i$ is supported on $\text{im}(M^T)$ and is given there by the Dirichlet kernel pushed forward through M^T .

This should not be confused with saying that conditioning is a pointwise Fourier multiplier on $\widehat{\mu}$. The algorithm multiplies by h_i in physical space, and therefore convolves in Fourier space. This convolution is exactly what produces the affine-coset average in Theorem 5.7.

5.2 Virtual-list distribution

Recall from Lemma 2.7 that for any probability distribution μ on G , if X, Y are independent with common law μ , then $X - Y$ has law

$$\nu := \mu * \check{\mu},$$

and

$$\widehat{\nu}(\omega) = |\widehat{\mu}(\omega)|^2.$$

Proposition 5.6. Let $\nu_i := \mu_{i-1} * \check{\mu}_{i-1}$ be the virtual-list distribution at level i . Then:

(i) $\widehat{\nu}_i(\omega) = |\widehat{\mu_{i-1}}(\omega)|^2$ for every $\omega \in G$.

(ii) ν_i is even.

(iii) If μ_{i-1} is even, then $\widehat{\mu_{i-1}}(\omega) \in \mathbb{R}$ for every ω , and hence

$$\widehat{\nu}_i(\omega) = \widehat{\mu_{i-1}}(\omega)^2.$$

In particular, for the empirical full-pair algorithm, part (i) is the correct formula at every level, including $i = 1$.

Proof. Part (i) is the convolution theorem plus Lemma 2.7.

For (ii), for every $x \in G$,

$$\nu_i(x) = \sum_{u \in G} \mu_{i-1}(u) \mu_{i-1}(u - x).$$

Also

$$\nu_i(-x) = \sum_{u \in G} \mu_{i-1}(u) \mu_{i-1}(u+x).$$

Substituting $v = u + x$ in the last sum gives

$$\nu_i(-x) = \sum_{v \in G} \mu_{i-1}(v-x) \mu_{i-1}(v) = \nu_i(x).$$

Thus ν_i is even.

Part (iii) follows from the final statement of Lemma 2.7. $\square$

5.3 Exact recurrence for $\widehat{\mu}_i$

We now combine the two operations. At level i , the algorithm draws a vector from ν_i and retains it if the conditioning filter h_i accepts. The unnormalized post-conditioning density is $g_i := \nu_i \cdot h_i$, and the normalized distribution is $\mu_i := g_i/\alpha_i$, where $\alpha_i := \sum_{x \in G} g_i(x)$ is the acceptance probability.

Theorem 5.7 (Exact Fourier recurrence). *For every $\omega \in G$,*

$$\widehat{g}_i(\omega) = \frac{1}{q^{b_i}} \sum_{\tau \in H} K_i(\tau) |\widehat{\mu_{i-1}}(\omega + A_i'^T \tau)|^2, \quad (20)$$

the acceptance probability is

$$\alpha_i = \widehat{g}_i(0) = \frac{1}{q^{b_i}} \sum_{\tau \in H} K_i(\tau) |\widehat{\mu_{i-1}}(A_i'^T \tau)|^2, \quad (21)$$

and the normalized recurrence is

$$\widehat{\mu}_i(\omega) = \frac{\sum_{\tau \in H} K_i(\tau) |\widehat{\mu_{i-1}}(\omega + A_i'^T \tau)|^2}{\sum_{\tau \in H} K_i(\tau) |\widehat{\mu_{i-1}}(A_i'^T \tau)|^2}. \quad (22)$$

If μ_{i-1} is even, then the modulus-squares in (20)–(22) may be replaced by ordinary squares.

Proof. Since $g_i = \nu_i \cdot h_i$ is a pointwise product, its Fourier transform is

$$\widehat{g}_i(\omega) = \sum_{x \in G} \nu_i(x) h_i(x) \chi_\omega(x).$$

Substitute the expansion (16) of h_i :

$$\widehat{g}_i(\omega) = \sum_{x \in G} \nu_i(x) \left(\frac{1}{q^{b_i}} \sum_{\tau \in H} K_i(\tau) \chi_{M^T \tau}(x) \right) \chi_\omega(x).$$

Interchanging finite sums,

$$\widehat{g}_i(\omega) = \frac{1}{q^{b_i}} \sum_{\tau \in H} K_i(\tau) \underbrace{\sum_{x \in G} \nu_i(x) \chi_{\omega + M^T \tau}(x)}_{=\widehat{\nu}_i(\omega + M^T \tau)}.$$

By Proposition 5.6(i),

$$\widehat{\nu}_i(\omega + M^T \tau) = |\widehat{\mu}_{i-1}(\omega + M^T \tau)|^2.$$

This proves (20). Setting $\omega = 0$ gives (21). The normalized recurrence follows from $\widehat{\mu}_i = \widehat{g}_i / \alpha_i$.

Finally, $\alpha_i > 0$ because ν_i assigns positive mass to 0 and $h_i(0) = \mathbf{1}_C(0) = 1$. The even case follows from Proposition 5.6(iii). $\square$

Remark 5.8 (Conditioning is not pointwise multiplication). The recurrence (22) shows that $\widehat{\mu}_i(\omega)$ depends on $\widehat{\mu}_{i-1}$ evaluated at all frequencies in the affine coset $\omega + \text{im}(A_i'^T)$, weighted by the kernel K_i . Schematically,

$$\widehat{\mu}_{i-1} \xrightarrow{\text{difference}} |\widehat{\mu}_{i-1}|^2 \xrightarrow{\text{condition on } \|-A_i'x\|_{\infty, q} \leq 2^i} \frac{(K_i *_{A_i'^T} |\widehat{\mu}_{i-1}|^2)(\omega)}{(K_i *_{A_i'^T} |\widehat{\mu}_{i-1}|^2)(0)},$$

where the pushforward convolution is

$$(K_i *_{A_i'^T} F)(\omega) := \sum_{\tau \in H} K_i(\tau) F(\omega + A_i'^T \tau).$$

This coupling between frequencies is the fundamental source of difficulty in the analysis.

Definition 5.9 (Denominator). We define the *level- i denominator* as

$$D_i := \sum_{\tau \in H} K_i(\tau) |\widehat{\mu}_{i-1}(A_i'^T \tau)|^2 = q^{b_i} \alpha_i. \quad (23)$$

Controlling D_i is one of the central technical challenges. Its exact analysis occupies Section 8.

6 The Affine-Goodness Framework

The recurrence (22) couples each output coefficient $\widehat{\mu}_i(\omega)$ to the input coefficients along the affine coset $\omega + \text{im}(A_i'^T)$. This section develops the correct notion of “good frequency” adapted to this structure.

Throughout, fix a level $i \geq 1$ and write

$$M := A_i' \in \mathbb{Z}_q^{b_i \times d}, \quad K := K_i, \quad \mu := \mu_{i-1}, \quad \mu^+ := \mu_i,$$

so that the recurrence reads

$$\widehat{\mu}^+(\omega) = \frac{\sum_{\tau \in H} K(\tau) |\widehat{\mu}(\omega + M^T \tau)|^2}{\sum_{\tau \in H} K(\tau) |\widehat{\mu}(M^T \tau)|^2}. \quad (24)$$

Define also

$$S := \sum_{\tau \in H} |K(\tau)|, \quad D := \sum_{\tau \in H} K(\tau) |\widehat{\mu}(M^T \tau)|^2, \quad \rho := \frac{D}{S}. \quad (25)$$

6.1 Affine-average goodness

For a threshold $\theta \in (0, 1)$, define the bad-frequency set

$$B(\theta) := \{\xi \in G : |\widehat{\mu}(\xi)| > \theta\}.$$

Thus

$$\delta(\theta) = \frac{|B(\theta)|}{q^d}.$$

Definition 6.1 (Normalized local bad mass). For $\omega \in G$ and $\theta \in (0, 1)$, define

$$m_\omega(\theta) := \frac{1}{S} \sum_{\tau \in H} |K(\tau)| \cdot \mathbf{1}_{B(\theta)}(\omega + M^T \tau).$$

Then $0 \leq m_\omega(\theta) \leq 1$.

Definition 6.2 (Affine-good frequency). Given $\theta \in (0, 1)$ and $\lambda \in [0, 1]$, call ω (θ, λ) -affine-good at level i if

$$m_\omega(\theta) \leq \lambda.$$

Proposition 6.3 (Coefficient bound for affine-good frequencies). *If ω is (θ, λ) -affine-good and $D > 0$, then*

$$|\widehat{\mu^+}(\omega)| \leq \frac{\theta^2 + (1 - \theta^2)\lambda}{\rho}, \quad \rho := D/S.$$

Proof. By the recurrence,

$$\widehat{\mu^+}(\omega) = \frac{\sum_{\tau \in H} K(\tau) |\widehat{\mu}(\omega + M^T \tau)|^2}{D}.$$

Since $D > 0$, the triangle inequality gives

$$|\widehat{\mu^+}(\omega)| \leq \frac{\sum_{\tau \in H} |K(\tau)| |\widehat{\mu}(\omega + M^T \tau)|^2}{D}.$$

Split the sum according to whether $\omega + M^T \tau \in B(\theta)$. On the complement of $B(\theta)$,

$$|\widehat{\mu}(\omega + M^T \tau)|^2 \leq \theta^2.$$

On $B(\theta)$, we use the trivial bound $|\widehat{\mu}(\omega + M^T \tau)|^2 \leq 1$. Therefore

$$\sum_{\tau \in H} |K(\tau)| |\widehat{\mu}(\omega + M^T \tau)|^2 \leq \theta^2 S + (1 - \theta^2) \sum_{\tau \in H} |K(\tau)| \mathbf{1}_{B(\theta)}(\omega + M^T \tau).$$

Since ω is (θ, λ) -affine-good, the final sum is at most λS . Hence

$$|\widehat{\mu^+}(\omega)| \leq \frac{\theta^2 S + (1 - \theta^2) \lambda S}{D} = \frac{\theta^2 + (1 - \theta^2) \lambda}{\rho}.$$

□

Proposition 6.4 (Mean local bad mass).

$$\frac{1}{q^d} \sum_{\omega \in G} m_\omega(\theta) = \delta(\theta). \tag{26}$$

Proof. By definition,

$$\frac{1}{q^d} \sum_{\omega \in G} m_{\omega}(\theta) = \frac{1}{Sq^d} \sum_{\omega \in G} \sum_{\tau \in H} |K(\tau)| \cdot \mathbf{1}_{B(\theta)}(\omega + M^T \tau).$$

Interchange the sums:

$$= \frac{1}{Sq^d} \sum_{\tau \in H} |K(\tau)| \underbrace{\sum_{\omega \in G} \mathbf{1}_{B(\theta)}(\omega + M^T \tau)}_{= |B(\theta)|},$$

where the inner sum equals $|B(\theta)|$ because $\omega \mapsto \omega + M^T \tau$ is a bijection of G for each fixed τ . Hence the expression equals $|B(\theta)|/(Sq^d) \cdot \sum_{\tau} |K(\tau)| = |B(\theta)|/q^d = \delta(\theta)$. $\square$

Remark 6.5. The identity (26) is exact and requires no assumption on the matrix $M = A'_i$. It relies only on the fact that translation is a bijection of G .

Corollary 6.6 (Majority of frequencies are affine-good). *For every $\lambda \in (0, 1]$,*

$$\frac{1}{q^d} \#\{\omega \in G : m_{\omega}(\theta) > \lambda\} \leq \frac{\delta(\theta)}{\lambda}. \quad (27)$$

Equivalently, at least a fraction $1 - \delta(\theta)/\lambda$ of frequencies are (θ, λ) -affine-good.

Proof. Apply Markov's inequality to the nonnegative random variable $m_{\omega}(\theta)$ under the uniform distribution on G , using $\mathbb{E}_{\omega}[m_{\omega}(\theta)] = \delta(\theta)$ from Proposition 6.4. $\square$

6.2 Recursive closure criterion

Combining Proposition 6.3 and Corollary 6.6 gives the exact criterion for the recursion to close.

Theorem 6.7 (Recursive closure criterion). *Fix parameters:*

- a previous-level threshold $\theta \in (0, 1)$,
- a local bad-mass tolerance $\lambda \in (0, 1]$,
- a normalization lower bound $\rho \geq \rho_* > 0$.

Define

$$\Theta^+(\theta, \lambda; \rho_*) := \frac{\theta^2 + (1 - \theta^2)\lambda}{\rho_*}.$$

Assume $\Theta^+(\theta, \lambda; \rho_) \in (0, 1)$. Then*

$$\delta(\mu^+, \Theta^+) \leq \frac{\delta(\mu, \theta)}{\lambda}.$$

Proof. By Corollary 6.6, at most a fraction $\delta(\theta)/\lambda$ of frequencies fail to be (θ, λ) -affine-good. For those that are, Proposition 6.3 gives $|\widehat{\mu^+}(\omega)| \leq \Theta^+(\theta, \lambda; \rho_*)$, since $\rho \geq \rho_*$. Hence $\delta(\mu^+, \Theta^+) \leq \delta(\theta)/\lambda$. $\square$

7 Deterministic Bounds on S_i

This section establishes tight bounds on the kernel ℓ^1 -mass $S_i = \sum_{\tau \in H_i} |K_i(\tau)|$.

7.1 The ℓ^1 -norm $T_q(W)$

Definition 7.1. For integers $W \geq 1$ and $q \geq 2$, set $B := (W - 1)/2$ (assuming W is odd) and define

$$T_q(W) := \sum_{a \in \mathbb{Z}_q} |D_B(a)|.$$

Since $|K_i(\tau)| = \prod_{j=1}^{b_i} |D_B(\tau_j)|$ with $B = 2^i$ and $W = W_i = 2B + 1$, the kernel mass factors as

$$S_i = T_q(W_i)^{b_i}. \quad (28)$$

Proposition 7.2 (Upper bound on $T_q(W)$). *There exists an absolute constant $C_+ > 0$ such that for all $1 \leq W < q$,*

$$T_q(W) \leq C_+ q (1 + \log W). \quad (29)$$

Proof. Set $B = (W - 1)/2$. By symmetry $|D_B(a)| = |D_B(-a)|$, so

$$T_q(W) \leq W + 2 \sum_{a=1}^{\lfloor q/2 \rfloor} \min\left(W, \frac{q}{2a}\right),$$

using Lemma 2.10(iv)–(v).

Let $A_0 := \lfloor q/(2W) \rfloor$.

If $A_0 \geq 1$, then

$$\sum_{a=1}^{A_0} \min\left(W, \frac{q}{2a}\right) \leq A_0 W \leq \frac{q}{2}.$$

Also,

$$\sum_{a=A_0+1}^{\lfloor q/2 \rfloor} \min\left(W, \frac{q}{2a}\right) \leq \frac{q}{2} \sum_{a=A_0+1}^{\lfloor q/2 \rfloor} \frac{1}{a} \leq \frac{q}{2} (1 + \log(q/2) - \log A_0).$$

Because $A_0 \geq 1$, it is guaranteed that $q/(2W) \geq 1$. If $q/(2W) \in [1, 2)$, then $\lfloor q/(2W) \rfloor = 1 > q/(4W)$. If $q/(2W) \geq 2$, then $\lfloor q/(2W) \rfloor > q/(2W) - 1 \geq q/(4W)$. So we have

$$\log(q/2) - \log A_0 \leq \log(2W),$$

so the second sum is at most $\frac{q}{2}(1 + \log(2W))$.

If $A_0 = 0$, then $W > q/2$, and therefore

$$\sum_{a=1}^{\lfloor q/2 \rfloor} \min\left(W, \frac{q}{2a}\right) \leq \frac{q}{2} \sum_{a=1}^{\lfloor q/2 \rfloor} \frac{1}{a} \leq \frac{q}{2} (1 + \log q).$$

Since $W > q/2$, one has $\log q \leq \log(2W)$.

Combining both cases gives

$$T_q(W) \leq C_+ q (1 + \log W)$$

for an absolute constant $C_+ > 0$. □

Proposition 7.3 (Lower bound on $T_q(W)$). *There exists an absolute constant $c_- > 0$ such that for all $3 \leq W \leq q/8$,*

$$T_q(W) \geq c_- q \log W. \quad (30)$$

Proof. Set $B = (W - 1)/2$. For each integer $0 \leq m \leq \lfloor (W - 1)/4 \rfloor$, define the interval

$$I_m := \left[\frac{(2m+1)q}{2W} - \frac{q}{8W}, \frac{(2m+1)q}{2W} + \frac{q}{8W} \right] \cap \mathbb{Z}.$$

Since $W \leq q/8$, the interval has length at least $q/(4W) \geq 2$, so $|I_m| \geq q/(8W)$.

For $a \in I_m$, the quantity $\pi W a/q$ lies within $\pi/8$ of $(2m+1)\pi/2$, so $|\sin(\pi W a/q)| \geq \cos(\pi/8) =: c_0 > 0$. Also $a < (m+1)q/W$, hence $\sin(\pi a/q) \leq \pi a/q \leq \pi(m+1)/W$. By (3),

$$|D_B(a)| = \left| \frac{\sin(\pi W a/q)}{\sin(\pi a/q)} \right| \geq \frac{c_0 W}{\pi(m+1)}.$$

Summing over $a \in I_m$,

$$\sum_{a \in I_m} |D_B(a)| \geq \frac{q}{8W} \cdot \frac{c_0 W}{\pi(m+1)} = \frac{c_0 q}{8\pi(m+1)}.$$

The intervals I_m are disjoint (successive centers are separated by $q/W > q/(8W)$ = the half-width, once $W \geq 3$). Summing over $0 \leq m \leq \lfloor (W - 1)/4 \rfloor$,

$$T_q(W) \geq \frac{c_0 q}{8\pi} \sum_{m=0}^{\lfloor (W-1)/4 \rfloor} \frac{1}{m+1} \geq \frac{c_0 q}{8\pi} \cdot c' \log W$$

for an absolute constant $c' > 0$, since the harmonic sum up to $\lfloor (W - 1)/4 \rfloor = \Theta(W)$ is $\Theta(\log W)$. $\square$

7.2 Consequence for S_i and the intrinsic loss

Corollary 7.4. *Under the parameter regime of Definition 3.10, for every level i such that $W_i \leq q/8$ (in particular, for every admissible level i), there exist absolute constants $c, C > 0$ such that*

$$(c q \log W_i)^{b_i} \leq S_i \leq (C q \log W_i)^{b_i}. \quad (31)$$

Proof. Combine (28) with Propositions 7.2 and 7.3, noting $W_i \geq 5 \geq 3$ and $\log W_i \geq \log 5 > 1$, so the $1 + \log W$ in (29) is $\Theta(\log W)$. $\square$

Definition 7.5 (Intrinsic normalization loss). The ratio between the geometric mass $|C_i| = W_i^{b_i}$ and the kernel ℓ^1 -mass S_i is

$$\frac{|C_i|}{S_i} = \left(\frac{W_i}{T_q(W_i)} \right)^{b_i} \asymp \left(\frac{W_i}{q \log W_i} \right)^{b_i}. \quad (32)$$

Since $W_i \leq 2\beta + 1 \ll q$ in the intended regime, this ratio is exponentially small in b_i . It represents the unavoidable cost of the signed Dirichlet kernel: even when the input distribution is perfectly uniform, the normalization ratio ρ_i can be no larger than $|C_i|/S_i$.

Definition 7.6 (Kernel constant). Let $C_{\ker} \geq 1$ be an absolute constant such that $T_q(W) \leq C_{\ker} q \log W$ for all $3 \leq W \leq q/8$. Such a constant exists by Proposition 7.2. Define the canonical normalization lower-bound scale

$$\rho_i := \frac{1}{2} \left(\frac{W_i}{C_{\ker} q \log W_i} \right)^{b_i}. \quad (33)$$

Under the admissibility conditions established in Lemmas 3.13 and 3.14,

$$\log \rho_i^{-1} = b_i \log \left(\frac{C_{\ker} q \log W_i}{W_i} \right) + O(1) = O \left(\frac{n}{\log \log n} \right). \quad (34)$$

8 Denominator Concentration via the Remainder R_i

This section gives a fully self-contained proof of denominator concentration that operates directly on the spectral recurrence and uses only the previous-level L^2 tail bound $\delta_{i-1}(\theta_{i-1})$ as input.

Throughout this section we fix a level $i \geq 1$ and write

$$\mathcal{F}_{i-1} := \sigma(L_0, A'_1, \dots, A'_{i-1}).$$

We condition on $\mathcal{F}_{i-1}$. Under this conditioning, the empirical distribution μ_{i-1} is fixed, while the fresh block

$$A'_i \in \mathbb{F}_q^{b_i \times d}$$

is uniform and independent of $\mathcal{F}_{i-1}$. We do not condition on A'_i being surjective or full row rank. The concentration argument below uses only the elementary fact that, for every fixed nonzero $\tau \in \mathbb{F}_q^{b_i}$, the vector $A_i'^T \tau$ is uniform in G .

8.1 Setup and decomposition

Recall from Definition 5.9 that

$$D_i = \sum_{\tau \in H_i} K_i(\tau) |\widehat{\mu_{i-1}}(A_i'^T \tau)|^2, \quad K_i(0) = W_i^{b_i}.$$

Since $\widehat{\mu_{i-1}}(0) = 1$, the $\tau = 0$ contribution is exactly $W_i^{b_i}$.

Definition 8.1 (Denominator remainder). The *level- i remainder* is

$$R_i := \sum_{\tau \in H_i^*} K_i(\tau) F(A_i'^T \tau), \quad H_i^* := H_i \setminus \{0\}, \quad (35)$$

where

$$F(\omega) := |\widehat{\mu_{i-1}}(\omega)|^2 \in [0, 1].$$

Thus $D_i = W_i^{b_i} + R_i$.

The signed mean of R_i is computed exactly, while its averaged absolute value and variance are bounded in terms of the previous-level tail data. This gives a one-sided denominator concentration estimate sufficient to lower-bound ρ_i .

Lemma 8.2 (Distribution of $A_i'^T \tau$ under uniform A'_i). *Under the conditional law of A'_i given $\mathcal{F}_{i-1}$:*

- (i) *For every fixed $\tau \in H_i^*$, $A_i'^T \tau \sim \text{Unif}(G)$.*
- (ii) *For every fixed $\tau, \sigma \in H_i^*$ that are $\mathbb{F}_q$ -linearly independent, $(A_i'^T \tau, A_i'^T \sigma) \sim \text{Unif}(G \times G)$.*
- (iii) *If $\sigma = a\tau$ for $a \in \mathbb{F}_q^\times$, then $A_i'^T \sigma = a A_i'^T \tau$.*

Proof. Write $A'_i = [a_1 \ \cdots \ a_d]$ column-wise, with the columns $a_k \in \mathbb{F}_q^{b_i}$ i.i.d. uniform. The k -th coordinate of $A_i'^T \tau$ is the linear form $\langle \tau, a_k \rangle$ on the random vector a_k . For $\tau \neq 0$, this linear form is nontrivial and hence uniform on $\mathbb{F}_q$; the coordinates are independent across k since the columns are. This proves (i). For (ii), the joint vector $(\langle \tau, a_k \rangle, \langle \sigma, a_k \rangle)$ is a nondegenerate linear function of $a_k \in \mathbb{F}_q^{b_i}$ when τ, σ are independent, hence uniform on $\mathbb{F}_q^2$, and joint independence across k gives the claim. Part (iii) is immediate from the linearity of $\tau \mapsto A_i'^T \tau$. $\square$

We use the following spectral proxy throughout.

Definition 8.3 (Spectral proxy). Set $\theta := \theta_{i-1} > 0$ and $\delta := \delta_{i-1}(\theta)$, $B_{i-1} := \{\omega \in G : |\widehat{\mu_{i-1}}(\omega)| > \theta\}$. Define

$$p_{i-1} := \theta^2 + (1 - \theta^2)\delta. \quad (36)$$

Then $0 \leq F(\omega) \leq 1$ for all ω , and the pointwise inequality

$$F(\omega) \leq \theta^2 + (1 - \theta^2)\mathbf{1}_{B_{i-1}}(\omega) \quad (37)$$

holds.

Justification of (37). On B_{i-1}^c , $|\widehat{\mu_{i-1}}(\omega)| \leq \theta$ so $F(\omega) \leq \theta^2$. On B_{i-1} , $F(\omega) \leq 1$. The right-hand side of (37) equals θ^2 on B_{i-1}^c and 1 on B_{i-1} ; in both cases it dominates F . $\square$

Averaging (37) over $\omega \sim \text{Unif}(G)$ gives

$$M := \frac{1}{q^d} \sum_{\omega \in G} F(\omega) \leq \theta^2 + (1 - \theta^2)\delta = p_{i-1}. \quad (38)$$

8.2 Mean of R_i

Lemma 8.4 (Kernel-sum identity). $\sum_{\tau \in H_i} K_i(\tau) = q^{b_i}$. Consequently, $\sum_{\tau \in H_i^*} K_i(\tau) = q^{b_i} - W_i^{b_i} > 0$.

Proof. Let $h := \mathbf{1}_{C_i} : H_i \rightarrow \mathbb{R}$. By Definitions 2.2 and 2.3, $\widehat{h}(\tau) = \sum_{y \in C_i} \chi_\tau(y) = \prod_{j=1}^{b_i} D_{2^i}(\tau_j) = K_i(\tau)$, using $\chi_\tau(y) = \prod_j e^{2\pi i \tau_j y_j / q}$ and the Dirichlet-kernel definition. Fourier inversion (2) on H_i at $0 \in C_i$ gives $1 = h(0) = q^{-b_i} \sum_{\tau} K_i(\tau)$, i.e. $\sum_{\tau} K_i(\tau) = q^{b_i}$. The second statement follows from $K_i(0) = W_i^{b_i}$. $\square$

Proposition 8.5 (Exact conditional mean).

$$\mathbb{E}[R_i \mid \mathcal{F}_{i-1}] = (q^{b_i} - W_i^{b_i}) M. \quad (39)$$

In particular,

$$0 \leq \mathbb{E}[R_i \mid \mathcal{F}_{i-1}] \leq (q^{b_i} - W_i^{b_i}) p_{i-1}. \quad (40)$$

Proof. By Lemma 8.2(i), for each fixed $\tau \in H_i^*$, $A_i'^T \tau \sim \text{Unif}(G)$, so

$$\mathbb{E}[F(A_i'^T \tau) \mid \mathcal{F}_{i-1}] = q^{-d} \sum_{\omega \in G} F(\omega) = M.$$

Linearity of expectation gives $\mathbb{E}[R_i \mid \mathcal{F}_{i-1}] = M \sum_{\tau \in H_i^*} K_i(\tau)$, and Lemma 8.4 evaluates the sum. The bounds in (40) follow from $0 \leq M \leq p_{i-1}$ (equation (38)) and $q^{b_i} > W_i^{b_i}$. $\square$

Remark 8.6. The sign of R_i is not constrained pointwise: individual values of $K_i(\tau)$ can be negative, and F is nonnegative, so the summands of R_i have mixed signs. What Proposition 8.5 asserts is that the *conditional mean is nonnegative*. This signed nonnegativity is what makes one-sided Chebyshev work for a lower bound on D_i in Section 8.4.

8.3 Variance via projective-line decomposition

Definition 8.7 (Line decomposition of R_i). For each line $\ell \in \mathbb{P}(H_i)$ (i.e. each one-dimensional $\mathbb{F}_q$ -subspace of H_i), set $\ell^* := \ell \setminus \{0\}$ and

$$Y_\ell := \sum_{\tau \in \ell^*} K_i(\tau) F(A_i'^T \tau), \quad L_i(\ell) := \sum_{\tau \in \ell^*} |K_i(\tau)|. \quad (41)$$

Since the lines $\ell \in \mathbb{P}(H_i)$ partition H_i^* ,

$$R_i = \sum_{\ell \in \mathbb{P}(H_i)} Y_\ell. \quad (42)$$

Lemma 8.8 (Pairwise independence across lines). *For distinct lines $\ell \neq \ell'$ in $\mathbb{P}(H_i)$, the random variables Y_ℓ and $Y_{\ell'}$ are independent given $\mathcal{F}_{i-1}$.*

Proof. Pick representatives $\tau_0 \in \ell^*$ and $\sigma_0 \in (\ell')^*$. Since $\ell \neq \ell'$, τ_0 and σ_0 are linearly independent, so by Lemma 8.2(ii), $X := A_i'^T \tau_0$ and $X' := A_i'^T \sigma_0$ are jointly uniform on $G \times G$, in particular independent. By Lemma 8.2(iii), $A_i'^T(a\tau_0) = aX$ and $A_i'^T(b\sigma_0) = bX'$ for $a, b \in \mathbb{F}_q^\times$. Hence Y_ℓ is a function of X alone and $Y_{\ell'}$ is a function of X' alone, so $Y_\ell \perp Y_{\ell'}$ given $\mathcal{F}_{i-1}$. $\square$

Corollary 8.9. $\text{Var}(R_i \mid \mathcal{F}_{i-1}) = \sum_{\ell \in \mathbb{P}(H_i)} \text{Var}(Y_\ell \mid \mathcal{F}_{i-1})$.

Proof. Pairwise independence (Lemma 8.8) suffices for additivity of variances over a finite sum. $\square$

Lemma 8.10 (Pairwise spectral correlation). *For $c \in \mathbb{F}_q^\times$, define*

$$\Lambda(c) := \frac{1}{q^d} \sum_{\eta \in G} F(\eta) F(c\eta).$$

Then for every $c \in \mathbb{F}_q^\times$,

$$\Lambda(c) \leq \theta^4 + 2\delta. \quad (43)$$

Proof. Define $E_c := B_{i-1} \cup c^{-1}B_{i-1}$. Since $\eta \mapsto c^{-1}\eta$ is a bijection of G , $|c^{-1}B_{i-1}| = |B_{i-1}| = \delta q^d$, and the union bound gives $|E_c| \leq 2\delta q^d$. For $\eta \notin E_c$, both $\eta \notin B_{i-1}$ and $c\eta \notin B_{i-1}$, so $F(\eta), F(c\eta) \leq \theta^2$ and $F(\eta)F(c\eta) \leq \theta^4$. For $\eta \in E_c$, use $F(\eta)F(c\eta) \leq 1$. Therefore

$$\Lambda(c) \leq \frac{1}{q^d} ((q^d - |E_c|)\theta^4 + |E_c|) \leq \theta^4 + \frac{|E_c|}{q^d} \leq \theta^4 + 2\delta. \quad \square$$

Proposition 8.11 (Per-line second moment). *For every $\ell \in \mathbb{P}(H_i)$,*

$$\mathbb{E}[Y_\ell^2 \mid \mathcal{F}_{i-1}] \leq (\theta^4 + 2\delta) L_i(\ell)^2. \quad (44)$$

Consequently $\text{Var}(Y_\ell \mid \mathcal{F}_{i-1}) \leq (\theta^4 + 2\delta) L_i(\ell)^2$.

Proof. Fix a representative $\tau_0 \in \ell^*$, so that $\ell^* = \{a\tau_0 : a \in \mathbb{F}_q^\times\}$. Set $X := A_i'^T \tau_0$; by Lemma 8.2(i), $X \sim \text{Unif}(G)$. By Lemma 8.2(iii), $A_i'^T(a\tau_0) = aX$ for $a \in \mathbb{F}_q^\times$, so

$$Y_\ell = \sum_{a \in \mathbb{F}_q^\times} K_i(a\tau_0) F(aX).$$

Squaring and taking the conditional expectation,

$$\mathbb{E}[Y_\ell^2 \mid \mathcal{F}_{i-1}] = \sum_{a,b \in \mathbb{F}_q^\times} K_i(a\tau_0) K_i(b\tau_0) \cdot q^{-d} \sum_{\omega \in G} F(a\omega) F(b\omega).$$

For each pair (a, b) , substitute $\eta = a\omega$ (a bijection of G):

$$q^{-d} \sum_{\omega \in G} F(a\omega) F(b\omega) = q^{-d} \sum_{\eta \in G} F(\eta) F((b/a)\eta) = \Lambda(b/a),$$

with $b/a \in \mathbb{F}_q^\times$. By Lemma 8.10, $\Lambda(b/a) \leq \theta^4 + 2\delta$ uniformly in (a, b) . Hence

$$\mathbb{E}[Y_\ell^2 \mid \mathcal{F}_{i-1}] \leq (\theta^4 + 2\delta) \sum_{a,b \in \mathbb{F}_q^\times} |K_i(a\tau_0)| |K_i(b\tau_0)| = (\theta^4 + 2\delta) L_i(\ell)^2,$$

by the definition of $L_i(\ell)$. The variance bound follows since $\text{Var}(Y_\ell \mid \mathcal{F}_{i-1}) \leq \mathbb{E}[Y_\ell^2 \mid \mathcal{F}_{i-1}]$. $\square$

Lemma 8.12 (Aggregate kernel ℓ^2 mass).

$$\sum_{\tau \in H_i} K_i(\tau)^2 = (qW_i)^{b_i}, \quad \sum_{\tau \in H_i^*} K_i(\tau)^2 = (qW_i)^{b_i} - W_i^{2b_i}. \quad (45)$$

Proof. Apply Parseval (Proposition 2.4) to the indicator $h := \mathbf{1}_{C_i}$ on H_i . As in Lemma 8.4, $\widehat{h}(\tau) = K_i(\tau)$. Parseval gives

$$\sum_{\tau \in H_i} K_i(\tau)^2 = q^{b_i} \sum_{y \in H_i} h(y)^2 = q^{b_i} |C_i| = q^{b_i} W_i^{b_i}.$$

Subtracting the $\tau = 0$ contribution $K_i(0)^2 = W_i^{2b_i}$ gives the second identity. $\square$

Theorem 8.13 (Variance of R_i).

$$\text{Var}(R_i \mid \mathcal{F}_{i-1}) \leq (q-1)((qW_i)^{b_i} - W_i^{2b_i})(\theta^4 + 2\delta) \leq q(qW_i)^{b_i}(\theta^4 + 2\delta). \quad (46)$$

Proof. By Corollary 8.9 and proposition 8.11,

$$\text{Var}(R_i \mid \mathcal{F}_{i-1}) \leq (\theta^4 + 2\delta) \sum_{\ell \in \mathbb{P}(H_i)} L_i(\ell)^2.$$

For each line ℓ , $|\ell^*| = q-1$, and Cauchy–Schwarz gives

$$L_i(\ell)^2 = \left(\sum_{\tau \in \ell^*} |K_i(\tau)| \right)^2 \leq (q-1) \sum_{\tau \in \ell^*} K_i(\tau)^2.$$

Summing over lines, since lines partition H_i^* ,

$$\sum_{\ell \in \mathbb{P}(H_i)} L_i(\ell)^2 \leq (q-1) \sum_{\tau \in H_i^*} K_i(\tau)^2 = (q-1)((qW_i)^{b_i} - W_i^{2b_i}),$$

by Lemma 8.12. This gives the first inequality in (46); the second uses $(qW_i)^{b_i} - W_i^{2b_i} \leq (qW_i)^{b_i}$ and $q-1 \leq q$. $\square$

8.4 Concentration of D_i via one-sided Chebyshev

Theorem 8.14 (Conditional concentration of D_i). *For every $\varepsilon \in (0, 1)$,*

$$\Pr\left[D_i < (1 - \varepsilon)W_i^{b_i} \mid \mathcal{F}_{i-1}\right] \leq \frac{q}{\varepsilon^2} \left(\frac{q}{W_i}\right)^{b_i} (\theta_{i-1}^4 + 2\delta_{i-1}). \quad (47)$$

Proof. Write $\mathbb{E}_*[\cdot] := \mathbb{E}[\cdot \mid \mathcal{F}_{i-1}]$ and similarly $\Pr_*$ and Var_* for brevity. By Definition 8.1, $D_i = W_i^{b_i} + R_i$. By Proposition 8.5, $\mathbb{E}_*[R_i] \geq 0$, hence

$$\{D_i < (1 - \varepsilon)W_i^{b_i}\} = \{R_i < -\varepsilon W_i^{b_i}\} \subseteq \{R_i - \mathbb{E}_*[R_i] < -\varepsilon W_i^{b_i}\} \subseteq \{|R_i - \mathbb{E}_*[R_i]| > \varepsilon W_i^{b_i}\},$$

where the first inclusion uses $\mathbb{E}_*[R_i] \geq 0$ and the second is trivial. Chebyshev's inequality yields

$$\Pr_*[D_i < (1 - \varepsilon)W_i^{b_i}] \leq \frac{\text{Var}_*(R_i)}{\varepsilon^2 W_i^{2b_i}}.$$

Substituting the variance bound from Theorem 8.13,

$$\Pr_*[D_i < (1 - \varepsilon)W_i^{b_i}] \leq \frac{q(qW_i)^{b_i}(\theta_{i-1}^4 + 2\delta_{i-1})}{\varepsilon^2 W_i^{2b_i}} = \frac{q}{\varepsilon^2} \left(\frac{q}{W_i}\right)^{b_i} (\theta_{i-1}^4 + 2\delta_{i-1}). \quad \square$$

Corollary 8.15 (Recursive lower bound on D_i). *Fix $\alpha > 1$ and $\Delta > 1$. Suppose, conditionally on $\mathcal{F}_{i-1}$, that*

$$\frac{\Delta^2}{(\Delta - 1)^2} q \left(\frac{q}{W_i}\right)^{b_i} (\theta_{i-1}^4 + 2\delta_{i-1}) < \frac{1}{\alpha}. \quad (48)$$

Then

$$\Pr\left[D_i \geq \frac{1}{\Delta} W_i^{b_i} \mid \mathcal{F}_{i-1}\right] \geq 1 - \frac{1}{\alpha}. \quad (49)$$

Proof. Apply Theorem 8.14 with $\varepsilon = 1 - 1/\Delta \in (0, 1)$, so that $(1 - \varepsilon)W_i^{b_i} = W_i^{b_i}/\Delta$. The right-hand side of (47) becomes

$$\frac{q}{(1 - 1/\Delta)^2} \left(\frac{q}{W_i}\right)^{b_i} (\theta_{i-1}^4 + 2\delta_{i-1}) = \frac{\Delta^2}{(\Delta - 1)^2} q \left(\frac{q}{W_i}\right)^{b_i} (\theta_{i-1}^4 + 2\delta_{i-1}) < \frac{1}{\alpha},$$

using (48). Hence $\Pr\left[D_i < \frac{1}{\Delta} W_i^{b_i} \mid \mathcal{F}_{i-1}\right] < 1/\alpha$, which is (49). $\square$

Looseness of the line decomposition. The factor $q - 1$ in (46) reflects the fact that all $q - 1$ nonzero scalar multiples of a fixed direction $\tau_0 \in H_i^*$ produce values $F(aX)$ that are deterministic functions of the single random vector $X = A_i'^T \tau_0$, and hence must be aggregated by Cauchy–Schwarz rather than treated as independent. A finer analysis of the signed line sums $\sum_{\tau \in \ell^*} K_i(\tau) F(A_i'^T \tau)$, exploiting character cancellation along the line, could in principle improve this factor, but the present bound is already sufficient for Corollary 8.15.

9 Recursive Propagation: Algorithm-Centered Analysis

This section gives an analysis built on the line-decomposition denominator concentration of Section 8 (specifically, Corollary 8.15). The recursion is parametrized by three free knobs per level – a denominator slack $\Delta_i > 1$, an affine-goodness tolerance $\lambda_i \in (0, 1]$, and a failure-budget allocation $a_i > 1$ – without committing to any particular asymptotic shape for the spectral tail, and is targeted directly at a quantitative success criterion for the algorithm.

9.1 Single-step transition

We separate the deterministic admissibility clause from the probabilistic good event.

Definition 9.1 (Admissible level). Say that level $i \geq 1$ is *admissible*, written $\mathbf{A}_i$, if

$$(A1) \quad W_i \leq q/8,$$

$$(A2) \quad b_i = o(d),$$

$$(A3) \quad b_i \log(q/W_i) = \Theta(n/\log \log n),$$

$$(A4) \quad b_i \log(C_{\ker} q \log W_i/W_i) = \Theta(n/\log \log n).$$

The recursion tracks a single deterministic pair $(\theta_i, \hat{\delta}_i)$ at each level, with the convention that $\hat{\delta}_i$ is a high-probability deterministic upper bound on the actual spectral tail $\delta_i(\theta_i)$.

Theorem 9.2 (Single-step transition). *Fix an admissible level $i \geq 1$ (Definition 9.1), and condition on $\mathcal{F}_{i-1} = \sigma(A'_1, \dots, A'_{i-1})$. Suppose $\delta_{i-1}(\theta_{i-1}) \leq \hat{\delta}_{i-1}$ for some $\theta_{i-1} \in (0, 1)$ and $\hat{\delta}_{i-1} \in (0, 1)$. Assume moreover that the value θ_i defined in (51) lies in $(0, 1)$. Choose parameters $\Delta_i > 1$, $\lambda_i \in (0, 1]$, $a_i > 1$ satisfying*

$$\frac{\Delta_i^2}{(\Delta_i - 1)^2} q \left(\frac{q}{W_i} \right)^{b_i} (\theta_{i-1}^4 + 2\hat{\delta}_{i-1}) < \frac{1}{a_i}, \quad (50)$$

and define

$$\theta_i := \Delta_i \left(\frac{C_{\ker} q \log W_i}{W_i} \right)^{b_i} [\theta_{i-1}^2 + (1 - \theta_{i-1}^2) \lambda_i], \quad \hat{\delta}_i := \frac{\hat{\delta}_{i-1}}{\lambda_i}. \quad (51)$$

Then with conditional probability at least $1 - 1/a_i$, $\delta_i(\theta_i) \leq \hat{\delta}_i$.

Proof. The left-hand side of (50) is monotone increasing in $\hat{\delta}_{i-1}$, so (50) also holds with the actual $\delta_{i-1}(\theta_{i-1}) \leq \hat{\delta}_{i-1}$ in place of $\hat{\delta}_{i-1}$. Corollary 8.15 with $\Delta = \Delta_i$, $\alpha = a_i$ then yields

$$\Pr \left[\rho_i \geq \frac{1}{\Delta_i} \left(\frac{W_i}{C_{\ker} q \log W_i} \right)^{b_i} \middle| \mathcal{F}_{i-1} \right] \geq 1 - \frac{1}{a_i},$$

using the deterministic kernel-mass upper bound $S_i \leq (C_{\ker} q \log W_i)^{b_i}$ from Corollary 7.4. On this event, the recursive closure criterion (Theorem 6.7) at level i , applied with input threshold θ_{i-1} , tolerance λ_i , and lower bound $\rho_* = \frac{1}{\Delta_i} (W_i / (C_{\ker} q \log W_i))^{b_i}$, produces $\Theta^+ = \theta_i$ as in (51) together with $\delta_i(\theta_i) \leq \delta_{i-1}(\theta_{i-1})/\lambda_i \leq \hat{\delta}_{i-1}/\lambda_i = \hat{\delta}_i$. $\square$

Remark 9.3 (Three knobs, one per-level inequality). The three parameters per level enter (50)–(51) differently:

- Δ_i scales the denominator slack: smaller Δ_i tightens the prefactor in (51) but enlarges $\Delta_i^2/(\Delta_i - 1)^2$ in (50).
- λ_i trades threshold inflation against bad-mass amplification: appearing additively in θ_i and inversely in $\hat{\delta}_i$.
- a_i allocates failure budget against side-condition tightness.

9.2 Spectral mass at the final level

The finite algorithm succeeds once the final indexed list L_r contains a nonzero vector. Since $\mu_r = \mu_{L_r}$ is the empirical distribution obtained by choosing a uniformly random final list position, the number of nonzero entries in L_r is exactly

$$|L_r| (1 - \mu_r(0)).$$

Thus any upper bound of the form $\mu_r(0) \leq (|L_r| - 1)/|L_r|$ implies the existence of a nonzero final entry. The next lemma gives a convenient Fourier-spectral sufficient condition for bounding $\mu_r(0)$.

Lemma 9.4 (Mass at zero from spectral data). *For any probability distribution μ on $G = \mathbb{Z}_q^d$ and any $\theta \in (0, 1)$,*

$$\mu(0) \leq \delta(\mu, \theta) + \theta.$$

Proof. Fourier inversion at $x = 0$ (Equation (2)) gives $\mu(0) = q^{-d} \sum_{\omega} \hat{\mu}(\omega) \leq q^{-d} \sum_{\omega} |\hat{\mu}(\omega)|$. Splitting the sum at threshold θ and using $|\hat{\mu}(\omega)| \leq 1$ on the bad set $B(\theta) = \{\omega : |\hat{\mu}(\omega)| > \theta\}$,

$$\mu(0) \leq \delta(\mu, \theta) \cdot 1 + (1 - \delta(\mu, \theta)) \cdot \theta \leq \delta(\mu, \theta) + \theta. \quad \square$$

9.3 End-to-end success

Theorem 9.5 (Finite-list success criterion for the recursive analysis). *Choose $\theta_0 \in (0, 1)$ and a deterministic number $\hat{\delta}_0 \in (0, 1)$. Let E_0 be an event, measurable with respect to L_0 , such that*

$$\Pr[E_0] \geq 1 - p_0 \quad \text{and} \quad E_0 \subseteq \{\delta_0(\theta_0) \leq \hat{\delta}_0\},$$

where δ_0 is computed from the empirical distribution $\mu_0 = \mu_{L_0}$.

Choose a per-level schedule $\{(\Delta_i, \lambda_i, a_i)\}_{i=1}^r$, and define $(\theta_i, \hat{\delta}_i)$ for $i = 1, \dots, r$ recursively by (51). Suppose:

- (C1) **Threshold admissibility.** For every $i = 0, \dots, r$, one has $0 < \theta_i < 1$.
- (C2) **Per-level admissibility.** For each $i \in \{1, \dots, r\}$, the side condition (50) holds with the recursively defined $(\theta_{i-1}, \hat{\delta}_{i-1})$.
- (C3) **Final mass.** $\hat{\delta}_r + \theta_r \leq \frac{1}{2}$.
- (C4) **Failure budget.** $p_0 + \sum_{i=1}^r 1/a_i \leq \frac{1}{2}$.

Then, with probability at least $\frac{1}{2}$ over the initial list L_0 and the random matrix blocks $A'_1, \dots, A'_r$, the final empirical distribution satisfies

$$\mu_{L_r}(0) \leq \frac{1}{2}.$$

Consequently, if $N_0 \geq 2$, then with probability at least $\frac{1}{2}$ the final list L_r contains a nonzero vector.

Proof. Let

$$E_i := \{\delta_i(\theta_i) \leq \hat{\delta}_i\}, \quad i = 1, \dots, r,$$

where δ_i is computed from the empirical law $\mu_i = \mu_{L_i}$.

By assumption, E_0 holds with probability at least $1 - p_0$ and implies $\delta_0(\theta_0) \leq \hat{\delta}_0$. Conditional on $\mathcal{F}_{i-1} = \sigma(L_0, A'_1, \dots, A'_{i-1})$ and on $E_0 \cap E_1 \cap \dots \cap E_{i-1}$, the hypothesis needed to apply Theorem 9.2 at level i is exactly (C2). Therefore

$$\Pr[E_i^c \mid \mathcal{F}_{i-1}] \leq \frac{1}{a_i}.$$

A union bound over the base event and all levels gives

$$\Pr \left[E_0 \cap \bigcap_{i=1}^r E_i \right] \geq 1 - p_0 - \sum_{i=1}^r \frac{1}{a_i} \geq \frac{1}{2}.$$

On this event, $\delta_r(\theta_r) \leq \hat{\delta}_r$, so

$$\delta_r(\theta_r) + \theta_r \leq \hat{\delta}_r + \theta_r \leq \frac{1}{2}$$

by (C3). Applying Lemma 9.4 to $\mu_r = \mu_{L_r}$ gives $\mu_{L_r}(0) \leq 1/2$.

It remains to translate this distributional statement into a finite-list statement. Since $N_0 \geq 2$ and every self-pair (s, s) is accepted at every level, all lists are nonempty and in fact $|L_r| \geq 2$. Hence Corollary 3.8 implies that L_r contains a nonzero entry. $\square$

Remark 9.6 (Choosing the base event). There are two useful choices of the base event.

For the population obstruction calculation, one may use the deterministic population estimate

$$\hat{\delta}_0 = \frac{1}{3^d \theta_0^2}.$$

This is not the finite empirical initialization, but it is stronger than the empirical bound and is therefore sufficient for deriving a negative result for the recurrence.

For the actual finite-list algorithm, take $N_0 = 3^r N$ and apply Lemma 4.8. With probability at least $1 - 1/\eta$ over L_0 , simultaneously for all $\theta \in (0, 1)$,

$$\delta_0(\theta) \leq \frac{\eta}{\theta^2} \left(\frac{1}{N_0} + \left(1 - \frac{1}{N_0} \right) 3^{-d} \right).$$

Thus for a fixed θ_0 one may set

$$p_0 := \frac{1}{\eta}, \quad \hat{\delta}_0 := \frac{\eta}{\theta_0^2} \left(\frac{1}{N_0} + \left(1 - \frac{1}{N_0} \right) 3^{-d} \right).$$

The higher-moment empirical bound Lemma 4.11 can be used analogously.

Remark 9.7 (Closed form and feasibility reduction). Iterating the second part of (51) from $\hat{\delta}_0 = \frac{1}{3^d \theta_0^2}$ gives the closed form

$$\hat{\delta}_r = \frac{1}{3^d \theta_0^2 \prod_{i=1}^r \lambda_i}, \tag{52}$$

while θ_r is obtained by unrolling the first part of (51). In the regime $\lambda_i \geq \theta_{i-1}^2$ at every level (the “Markov-dominated” regime), the threshold update simplifies to

$$\theta_i \leq 2\Delta_i \left(\frac{C_{\ker} q \log W_i}{W_i} \right)^{b_i} \lambda_i. \tag{53}$$

Theorem 9.5 thus reduces the analysis of algorithmic success to the existence of a parameter schedule $(\theta_0, \{(\Delta_i, \lambda_i, a_i)\}_{i=1}^r)$ simultaneously satisfying (C2), (C3), (C4) under Definition 3.10. The constraints couple the cumulative bad-mass amplification $(\prod_i \lambda_i)^{-1}$, the per-level threshold inflation $\Delta_i (C_{\ker} q \log W_i / W_i)^{b_i}$, and the failure-budget allocation $1/a_i$ across $r = \Theta(\log n)$ levels.

9.4 An obstruction within the affine-goodness recurrence

This subsection establishes a negative result complementing the analysis of §9.1–§9.3. Within the recurrence (51) of Theorem 9.2, the final-mass condition (C3) alone forces

$$d = \Omega(n \log n / \log \log n).$$

Thus this affine-goodness recurrence cannot prove success under the sole dimension hypothesis $d = \omega(n / \log \log n)$; it requires a strictly stronger dimension scale.

Definition 9.8. At each admissible level i , define

$$T_i := \left(\frac{C_{\ker} q \log W_i}{W_i} \right)^{b_i}. \quad (54)$$

By Lemma 3.14, $\log T_i = \Theta(n / \log \log n)$ for every admissible level i .

Lemma 9.9 (Lower bound on cumulative amplification). *Let $\{(\Delta_i, \lambda_i)\}_{i=1}^r$ be any parameter schedule with $\Delta_i \geq 1$ and $\lambda_i \in (0, 1]$, applied through (51) from some $\theta_0 \in (0, 1)$, and suppose the resulting recursive thresholds satisfy $\theta_i < 1$ for every $i \in \{1, \dots, r\}$. Then for n sufficiently large,*

$$\sum_{i=1}^r \log(1/\lambda_i) \geq \sum_{i=1}^r \log T_i - r \log 2 = \Omega\left(\frac{n \log n}{\log \log n}\right). \quad (55)$$

Proof. Fix $i \in \{1, \dots, r\}$. The hypothesis $\theta_i < 1$ together with (51) gives

$$\Delta_i T_i [\theta_{i-1}^2 + (1 - \theta_{i-1}^2) \lambda_i] < 1.$$

Since $\theta_{i-1} < 1$ (by hypothesis for $i \geq 2$, by assumption for $i = 1$), both summands inside the bracket are nonnegative, so each is bounded by $1/(\Delta_i T_i)$. In particular, using $\Delta_i \geq 1$,

$$\theta_{i-1}^2 < \frac{1}{\Delta_i T_i} \leq \frac{1}{T_i}. \quad (56)$$

By Lemma 3.14, $\log T_i = \Theta(n / \log \log n) \rightarrow \infty$, so $T_i \rightarrow \infty$ uniformly in i . Hence for n sufficiently large, (56) gives $\theta_{i-1}^2 \leq 1/2$ at every level, and consequently $1 - \theta_{i-1}^2 \geq 1/2$. The bracket inequality then yields

$$\Delta_i T_i (1 - \theta_{i-1}^2) \lambda_i < 1,$$

so

$$\lambda_i \leq \frac{1}{\Delta_i T_i (1 - \theta_{i-1}^2)} \leq \frac{2}{\Delta_i T_i} \leq \frac{2}{T_i}.$$

Taking logarithms, $\log(1/\lambda_i) \geq \log T_i - \log 2$. Summing over $i = 1, \dots, r$ and invoking Lemma 3.14 together with $r = \Theta(\log n)$,

$$\sum_{i=1}^r \log(1/\lambda_i) \geq \sum_{i=1}^r \log T_i - r \log 2 = \Theta\left(\frac{rn}{\log \log n}\right) - O(\log n) = \Omega\left(\frac{n \log n}{\log \log n}\right). \quad \square$$

Theorem 9.10 (Obstruction within the affine-goodness recurrence). *Let $\theta_0 \in (0, 1)$ and $\{(\Delta_i, \lambda_i)\}_{i=1}^r$ be any parameter schedule with $\Delta_i \geq 1$ and $\lambda_i \in (0, 1]$, applied through (51). If the resulting (θ_r, δ_r) satisfies (C3) of Theorem 9.5, then for n sufficiently large,*

$$d \log 3 \geq 2|\log \theta_0| + \sum_{i=1}^r \log T_i - r \log 2 + \log 2. \quad (57)$$

In particular,

$$d = \Omega\left(\frac{n \log n}{\log \log n}\right). \quad (58)$$

Proof. Step 1: (C3) forces $\theta_i < 1$ for every $i \in \{1, \dots, r\}$. Since (C3) gives $\theta_r \leq 1/2$, it is already impossible that $\theta_r \geq 1$. Suppose for contradiction that $\theta_j \geq 1$ for some $j < r$, and let j be the smallest such index. Then $\theta_{j-1} < 1$ (with $\theta_0 < 1$ by assumption, and $\theta_{j-1} < 1$ for $j \geq 2$ by minimality). For any $\lambda_{j+1} \in (0, 1]$,

$$\theta_j^2 + (1 - \theta_j^2)\lambda_{j+1} = \lambda_{j+1} + \theta_j^2(1 - \lambda_{j+1}) \geq \lambda_{j+1} + (1 - \lambda_{j+1}) = 1,$$

using $\theta_j^2 \geq 1$ and $1 - \lambda_{j+1} \geq 0$. Hence

$$\theta_{j+1} \geq \Delta_{j+1} T_{j+1} \geq T_{j+1}.$$

Since $T_i \rightarrow \infty$ uniformly over admissible levels (Definition 9.8), for n sufficiently large this implies $\theta_{j+1} > 1$. Repeating the argument gives $\theta_k \geq T_k > 1$ for every $k = j + 1, \dots, r$, contradicting $\theta_r \leq 1/2$.

Step 2: dimension lower bound. By Step 1 the hypothesis of Lemma 9.9 holds, so

$$\sum_{i=1}^r \log(1/\lambda_i) \geq \sum_{i=1}^r \log T_i - r \log 2.$$

The closed form (52) together with $\hat{\delta}_r \leq 1/2$ from (C3) gives $3^d \theta_0^2 \prod_{i=1}^r \lambda_i \geq 2$, hence

$$d \log 3 \geq 2|\log \theta_0| + \sum_{i=1}^r \log(1/\lambda_i) + \log 2 \geq 2|\log \theta_0| + \sum_{i=1}^r \log T_i - r \log 2 + \log 2,$$

which is (57). Since $r = \Theta(\log n)$ and $\log T_i = \Theta(n/\log \log n)$, (58) follows. $\square$

Corollary 9.11 (Insufficiency of the parameter regime). *Under Definition 3.10 with the additional restriction $d = o(n \log n / \log \log n)$, no parameter schedule $(\theta_0, \{(\Delta_i, \lambda_i, a_i)\}_{i=1}^r)$ with $\Delta_i \geq 1$ and $\lambda_i \in (0, 1]$ satisfies all of (C2)–(C4) of Theorem 9.5.*

Proof. By Theorem 9.10, satisfying (C3) alone forces $d = \Omega(n \log n / \log \log n)$, which is incompatible with $d = o(n \log n / \log \log n)$. $\square$

Remark 9.12 (Scope of the obstruction). Theorem 9.10 is a statement about the analytical route developed in this paper, not about Wagner’s algorithm itself. Any analysis that

- passes through the affine-goodness criterion (Theorem 6.7) for spectral propagation,
- uses the line-decomposition denominator concentration of Corollary 8.15 (or any analogue producing a per-level amplification $\hat{\delta}_i = \hat{\delta}_{i-1}/\lambda_i$ with $\lambda_i \in (0, 1]$),
- iterates the resulting one-step bound across all $r = \Theta(\log n)$ levels,

must spend at least $\Theta(n \log n / \log \log n)$ of the budget $d \log 3$ on cumulative bad-mass amplification. The mechanism is the per-level inflation factor T_i in (51): keeping $\theta_i < 1$ at each level forces $\lambda_i \leq 2/T_i$, and these multiply across levels to a $\hat{\delta}_r$ amplification of order

$$\prod_{i=1}^r (T_i/2) = 2^{-r} \prod_{i=1}^r T_i.$$

Whether an alternative analytical strategy—one that does not iterate (51) level-by-level, or that tracks a finer spectral statistic than $\delta_i(\theta_i)$ —can establish the success of the algorithm under the original Definition 3.10 is left open.

Interpretation. The obstruction above should be read as a limitation of the particular one-parameter spectral-tail recursion developed in this paper. The exact Fourier recurrence remains valid for the full-pair process, and the denominator concentration mechanism gives a rigorous way to lower-bound the normalization. What fails is the attempt to propagate only the scalar statistic $\delta_i(\theta_i)$ through affine-goodness and Markov amplification at each level. A successful analysis of the original Wagner heuristic, if one exists without discrete-Gaussian modifications, likely has to track more structure: for example, the geometry of the large Fourier coefficients, cancellation in the signed Dirichlet sums, or correlations between neighboring affine cosets.

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